





CNC Machine Tool Programming

seen main machine tool features + auxiliary devices
PROGRAM needed to use all kind of {NC} features (numerical control)
(act so that we can obtain NC machining operation)

Groover M.P., Automation, Production Systems, and Computer-Integrated Manufacturing: Chapter 7



The student knows:

- How to program CNC machine tools;
- What the benefits of CAD/CAM integration are.

Logic on how we program machine tool + **main language** to program it!
(BASICS → to plan industrial device motion)

Basic of Language → evolution (avoid time-consuming + error in machine)
↓
SW tools to optimize



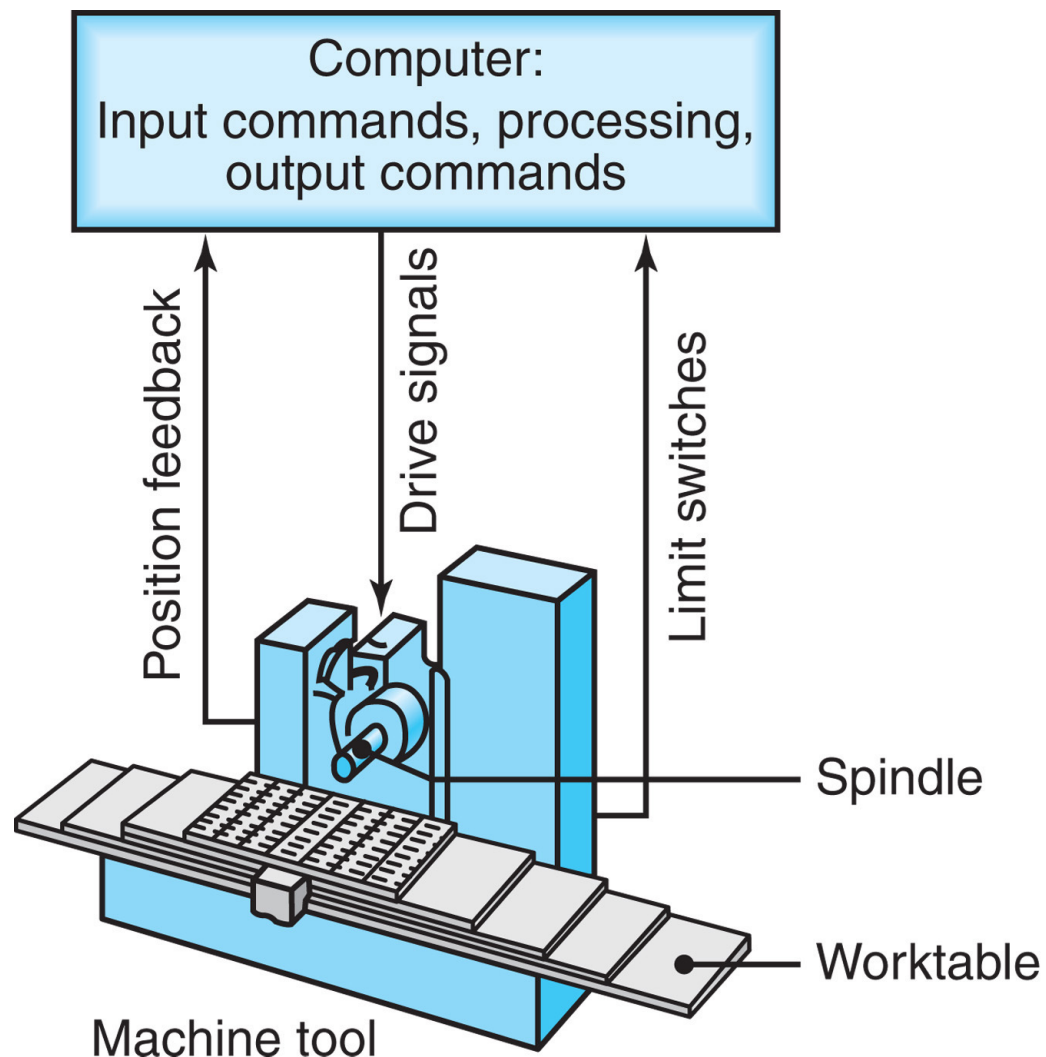


Numerical Control

Automatic control of a process performed by a device that makes use of numerical data introduced while the operation is in progress (ISO 2806:1994)

Computer Numerical Control

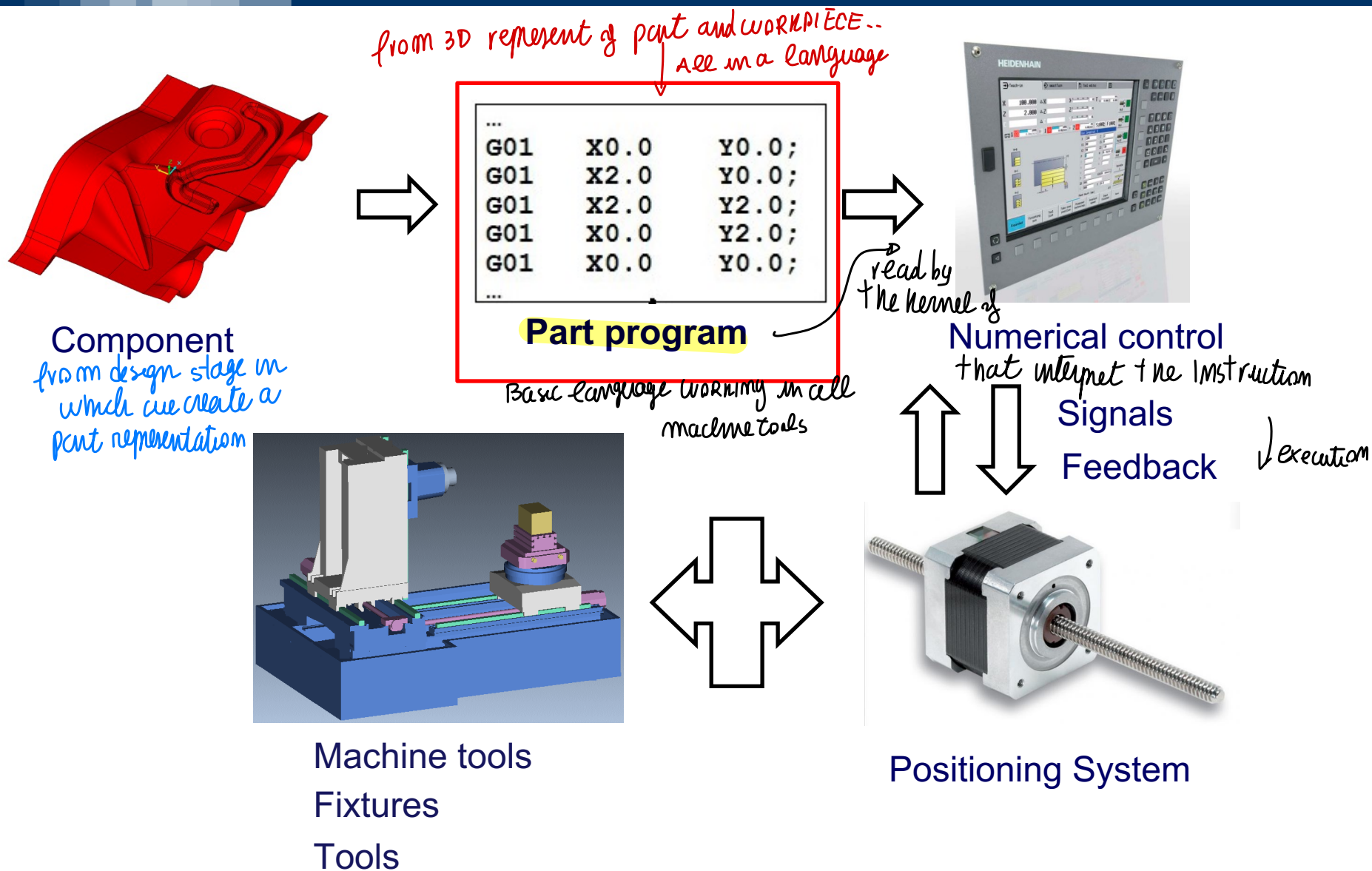
Realization of NC using a computer to control the machine functions (ISO 2806:1994)





MACHINE TOOL PROGRAMMING

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Ordered set of instructions in a language and in a format required to cause operations to be effected under automatic control, which is either written in the form of a **machine program** on an input medium or prepared as input data for processing in a computer to obtain a machine program (ISO 2806:1994).

↑
we can use maching
interface to directly program

old standard but still
confermed and used

Language to perform operation ⇒



PART PROGRAM: ISO CODE

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- NC commands such as G, M, S, T, H, D, F code and related address
- Call of sub-program and displaying message
- Setting variable and conditional program calls

† typical
language composed
of LINES,
lines=block
↑
elementary
operation
↓
many commands
in block
↓
expressed by letters
each with meaning
3 digit code
Letter Number Number
to give INFORM.

P123456 ; —————→ Program number

N10 G90 ; —————→ EOB(End of Block) code

N20 (G01) X50 Y20 F120 S100 ; —————→ Word

N30 X15 ;

N40 (Y)20 M03 ; —————→ Address

N50 G04 P100 ; —————→ Number

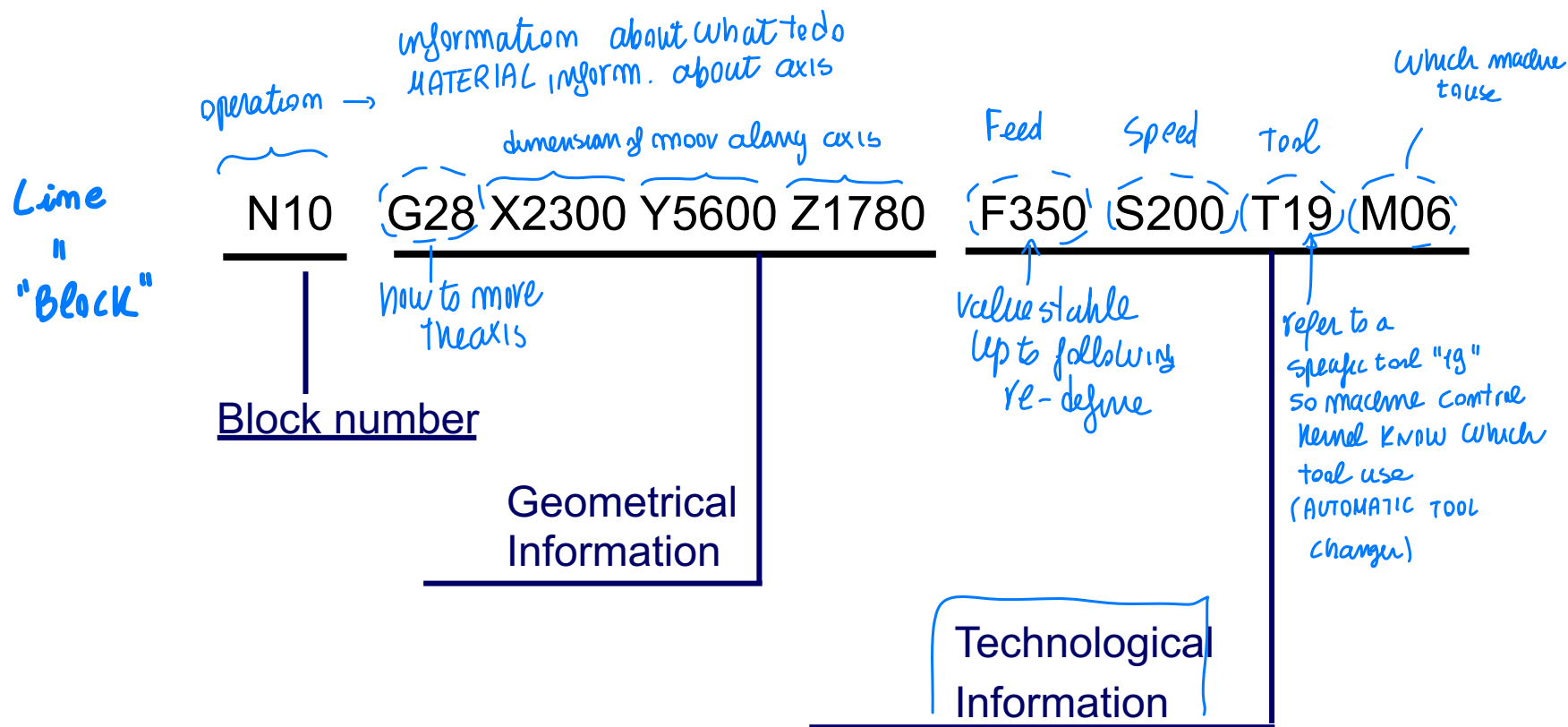
(N60) M30 ; —————→ Block number

Block ←



BLOCK EXAMPLE

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Motion *↘ different kind of letter address*

A, B, C, U, V, W, X, Y and Z

Addresses

Machine functions

F, G, M, N, S and T

types of speed--

A	B	C	D	E	F	G	H	I	L	M	N
O	P	Q	R	S	T	U	V	W	X	Y	Z

Modal: effective throughout the following blocks until the modal cancel command is used. *(valid until next change)*

Non-modal: effective within the commanded block and automatically cancelled by the next block.
(valid until next block)



MOTION ADDRESSES

→ Related to configuration

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X main motion of X linear axis

Y main motion of Y linear axis

Z main motion of Z linear axis

A angular coordinate around X

B angular coordinate around Y

C angular coordinate around Z

ROTATIONS

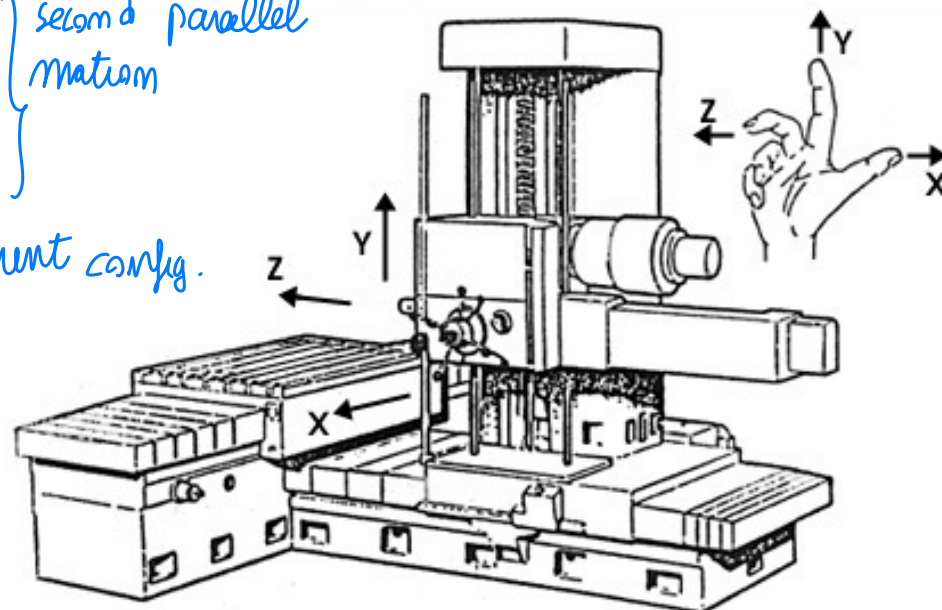
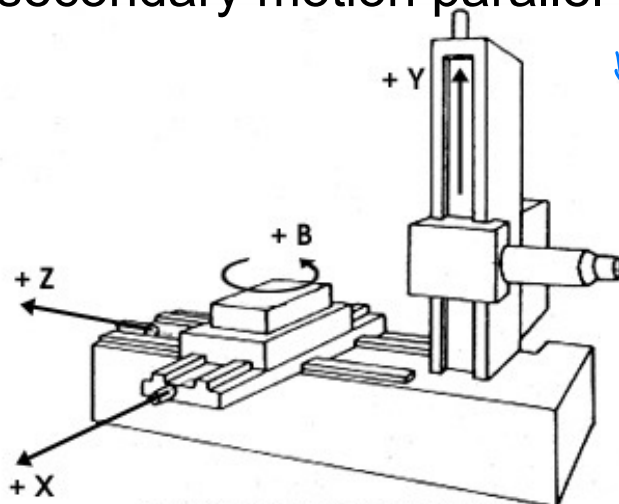
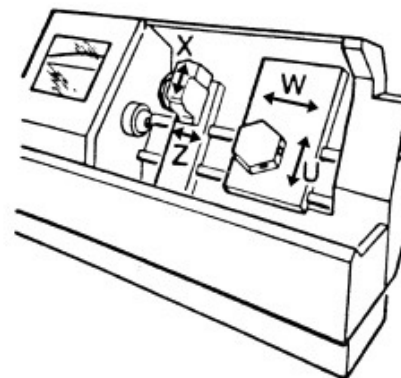
U secondary motion parallel to X

V secondary motion parallel to Y

W secondary motion parallel to Z

second parallel motion

↓ different config.





COORDINATE SYSTEMS

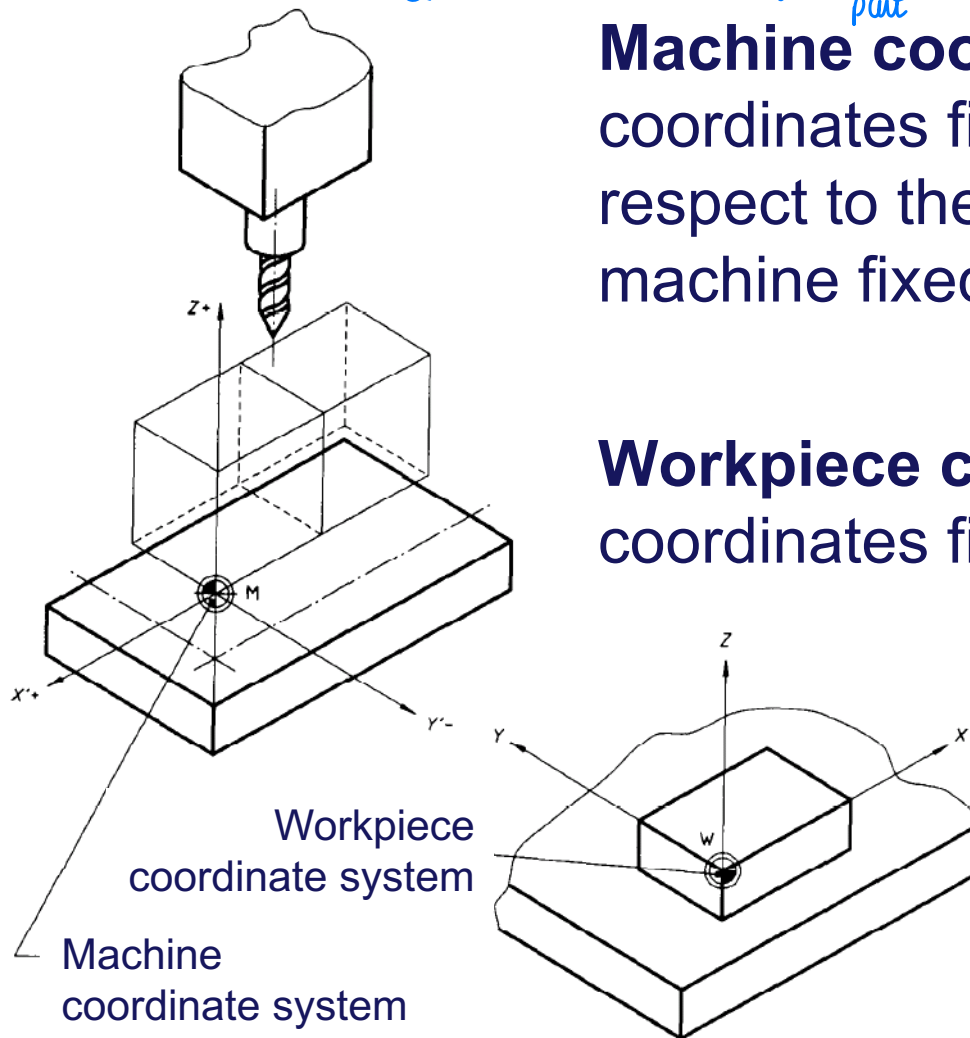
10

to define how the machine has
to machine the part

describe motion looking to the part
you should use reference

Machine coordinate system: Cartesian coordinates fixed on a machine with respect to the machine zero (origin of the machine fixed by the manufacturer).

Workpiece coordinate system: Cartesian coordinates fixed on the workpiece.



easily find ref syst. and
refer to this for programming

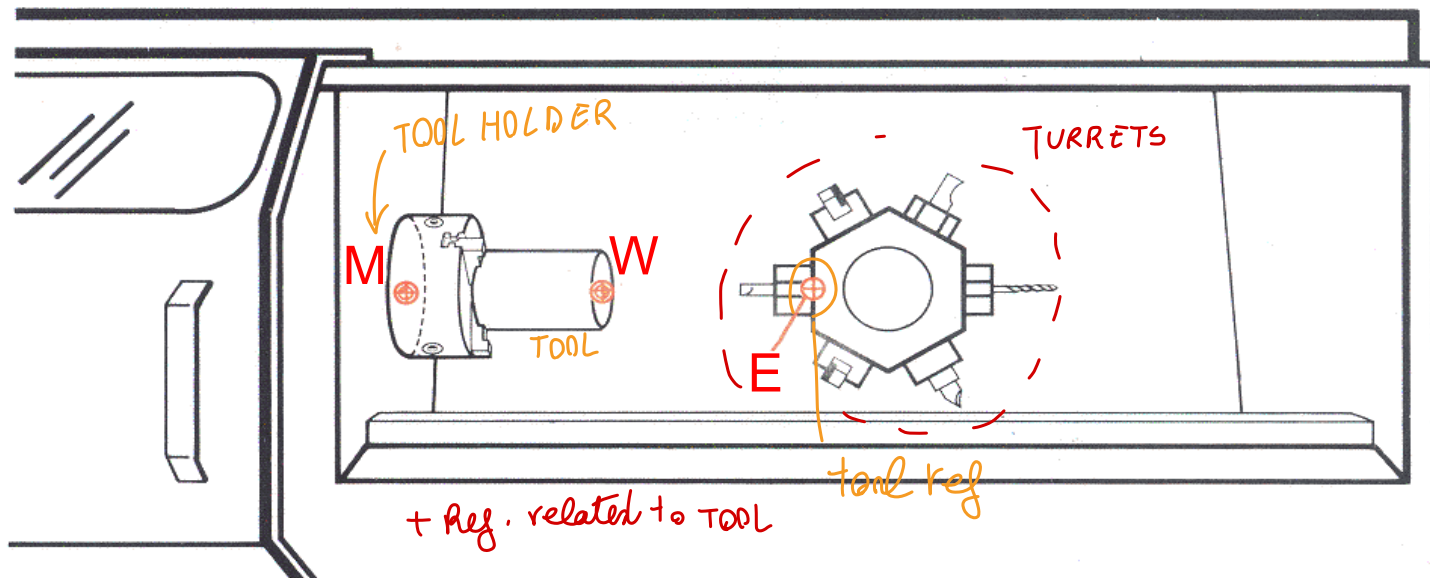
MOVE the tool made WORK ENVELOP..
position respect machine ref -
we have to know where is the part

we may have command for measurement respect reference → math alignment done
by CMM or machine tool



REFERENCE POINTS

We need to know the axis of Ref of machine tool → define WORKPIECE



- **M**: machine coordinate origin (machine zero)
- **W**: workpiece coordinate origin (part zero)

- **E**: tool coordinate origin (tool zero)

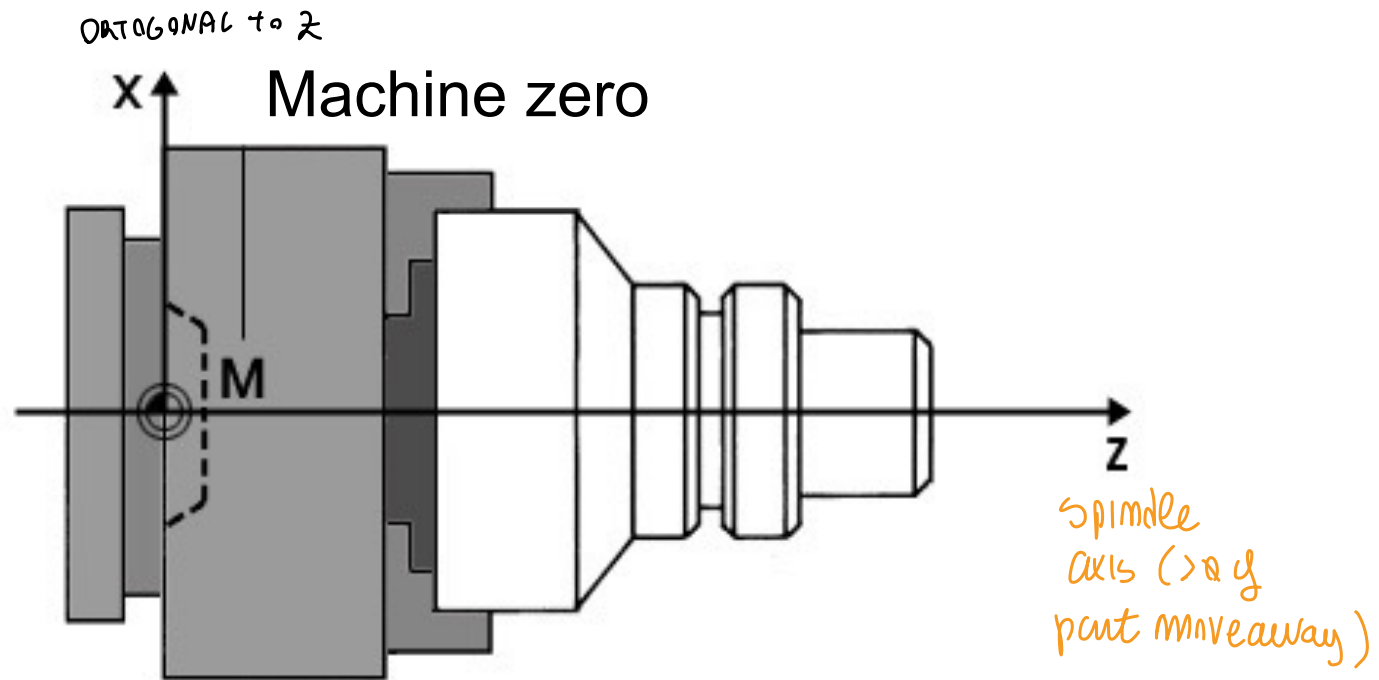
Coincident Ref to position the TOOL head

• ORIGIN
• AXIS
Ref syst } known!



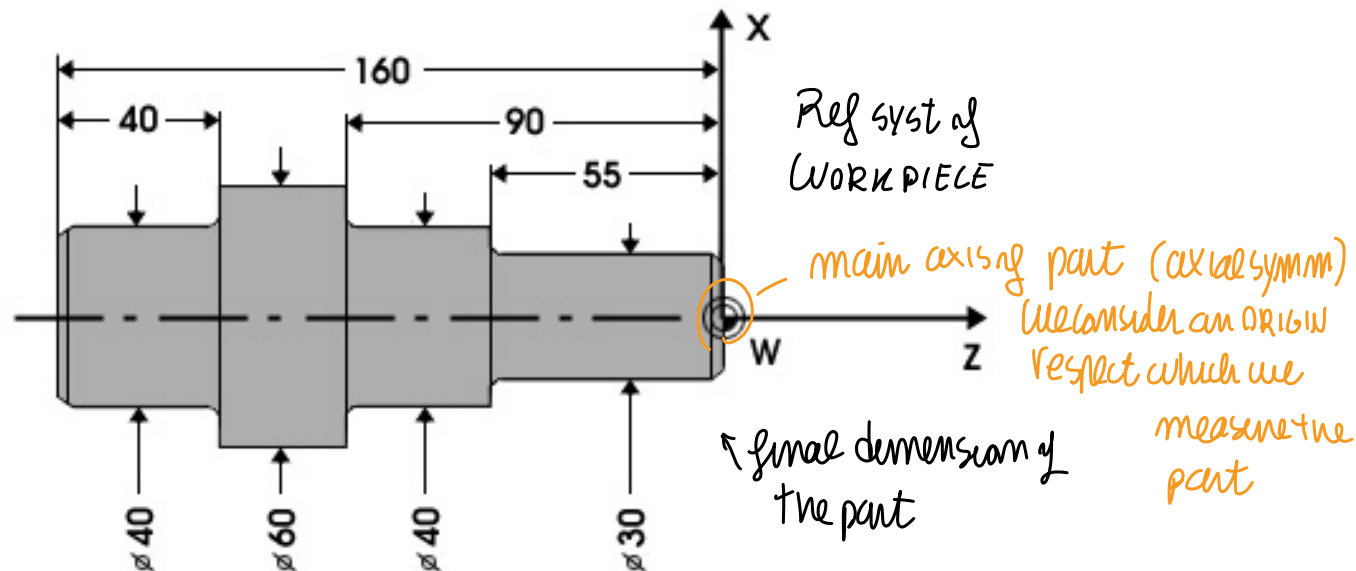
MACHINE ZERO IN TURNING CENTERS

POSITION the part on a lathe turning operation



Selected on the basis of:

- Technical drawing
- Convenience of setup and operation
- Safety of working conditions





PREPARATORY FUNCTIONS (G-CODE) *after local ref settled*

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We need the commands knowledge for movement (feed motion)

- **Movement:** commands to define the relative movement between tool and workpiece.
- **Setting local coordinate system:** commands to specify the origin and orientation of local coordinate system including orthogonal coordinate system, the polar coordinate system, and rotational coordinate system.
- **Interpolation:** commands to cut various profiles such as line, arc, helical, spline, etc. *(how to interpolate motion!)*
- **Miscellaneous:** commands to define tool compensation, safety check, etc. *enables you to compensate (through correctors) correct position to have correct machining*

SAFETY ISSUE in

machining (like closed machine) → closure able to support HIGH STRESS! (if broken piece of tool while rotating closure should resist impacts!) → This must be checked by a control system



AUXILIARY FUNCTIONS (M-CODE)

- **Coolant**: coolant on or off.
- **Start/End** of a main and sub- program: beginning and ending of a part program.
- **Spindle**: commands to let the spindle to rotate in CCW or CW direction and specify the spindle speed, limit speed and spindle gear change.
- **Miscellaneous**: commands to change the tool, the pallet, etc. *others enable changing --*



M COMMAND

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It can be the only instruction in the block, three (3) is the limit on the number of M instructions in a block.

It is activated before the movement commanded in the same block (G instruction).



N10 G00 X-60 Y-20 S500 M03 M08
└──────────┘
movement



M COMMAND

{COMMANDS for all OPERATIONS}

machine functions, different numbers

different instruction

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M00 stop program

M03 CW spindle rotation *clockwise*

M04 CCW spindle rotation *counterclockwise*

M05 Stop spindle rotation

M06 Automatic Tool Change → *start to change the tool
(to combine by an address
that identifies the tool)*

M08 Coolant on

M09 Coolant off

M72 Tailstock block *axis move*

M10/11 Chuck open/close (lathe) ← *chuck control*

M60 Automatic Pallet Change

M30 End of the program

... ↑ *some are defined by STANDARDS + some command out of ISO-code → specific changes of
language taking advantage of control syst. (dialect)*



SPINDLE CONTROL (S COMMAND) *basic spindle control example*

- The address S relates to machine spindle function
- Two alternatives:
 - Cutting speed in [m/min] – G96 *try to define a $V_c \approx \text{constant}$*
 - Spindle speed in [rpm] – G97 *if define speed as RPM we maintain rotation but V_c VARY*
- The spindle rotation direction must be specified: *value of spindle Rot*
 - Clockwise (CW) – M03
 - Counterclockwise (CCW) – M04
- Spindle will not start rotating until both the speed and the rotation commands have been processed
- Stop the spindle rotation – M05
- Other M-commands stop the rotation (M00, M01, M02, M30, etc.)



FEED RATE CONTROL (F COMMAND)

similarly to define the feed!

- The address **F** relates to how fast the tool will move
- Two alternatives:
 - Feed rate per minute [mm/min] – G94
 - Feed rate per revolution [mm/rev] - G95
(VELOCITY)
- Maximum programmable cutting feed rate is determined by machine manufacturer *≠ CONTROL SYST.*
- While running a program, the programmed feed rate may be temporarily suspended or changed by *feedhold override switch* and *feed rate override switch*



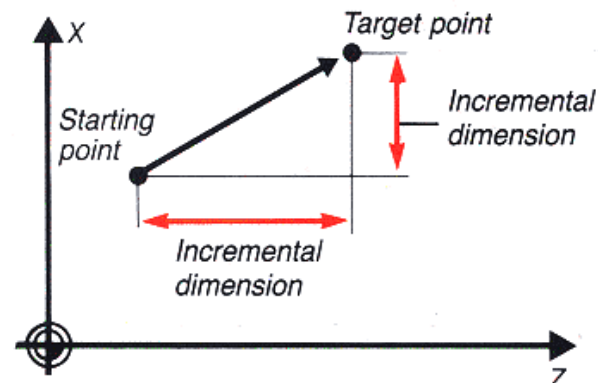
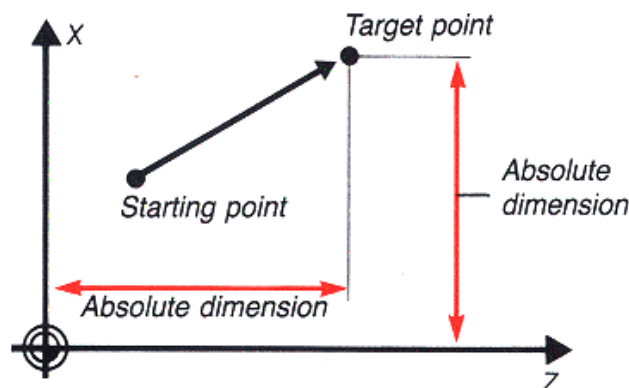
TYPES OF COORDINATES

to speak about motion
↓ how to define Ref. syst

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- G90: absolute coordinate system
 - G91: incremental coordinate system
- describe motion inside WORKPIECE ref

we should define if we
are giving command
absolute or incremental!



usually we give TARGET point of the motion
from previous point → FINAL POINT to reach



COORDINATES IN TURNING

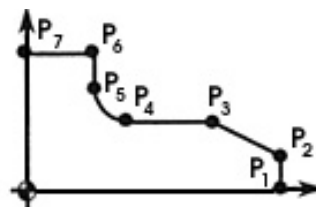
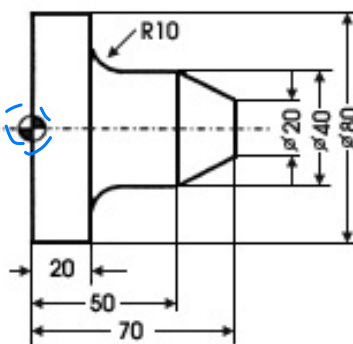
21

In turning absolute quotes are on the diameter,
incremental on the radius.

you can define the profile of the part to describe the motion in different ways--

*Ref
syst*

G90 Absolute coordinates

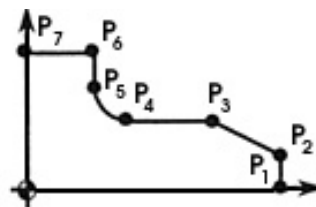
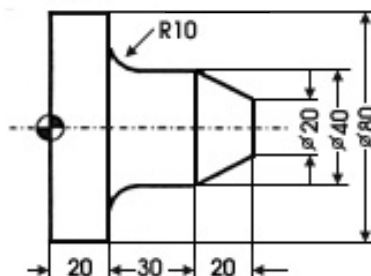


	X	Z
P1	0	70
P2	20	70
P3	40	50
P4	40	30
P5	60	20
P6	80	20
P7	80	0

absolute respect Ref.

Some points defined in different ways!

G91 Incremental coordinates



	X	Z
P1	0	0
P2	10	0
P3	10	-20
P4	0	-20
P5	10	-10
P6	10	0
P7	0	-20

We refer directly to the diameter from 2 D axial symm part, insert diam to avoid errors when programming!

always a reason to use a language in a given way -- directly programming!

*! In TURNING when refer to absolute val you refer to Diameter in that point
incremental value you give the Δ value difference--*



RAPID POSITIONING (G00)

MOTION

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once defined how to measure → MOVE the tool

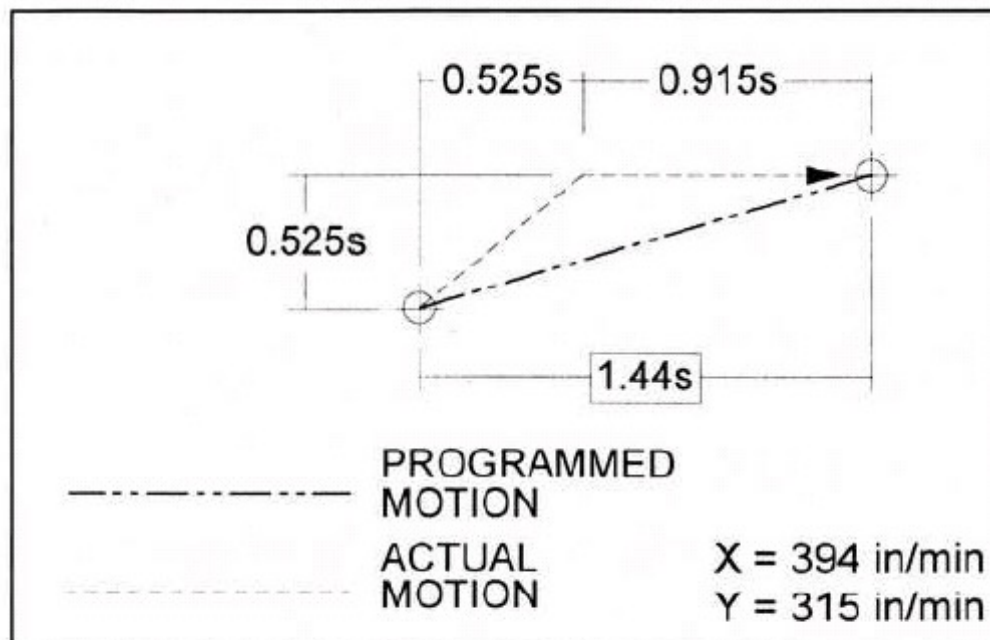
The rapid movement function (**G00**) is used for commanding the specific axes to **move rapidly** to the programmed position.

from INITIAL to FINAL position moving
Independently the axis! → MOVE FASTER

each axis --

this can be a
complex traj,
so we need free space
(safe condition)

always safe points
to be able to rapidly move
the tool,



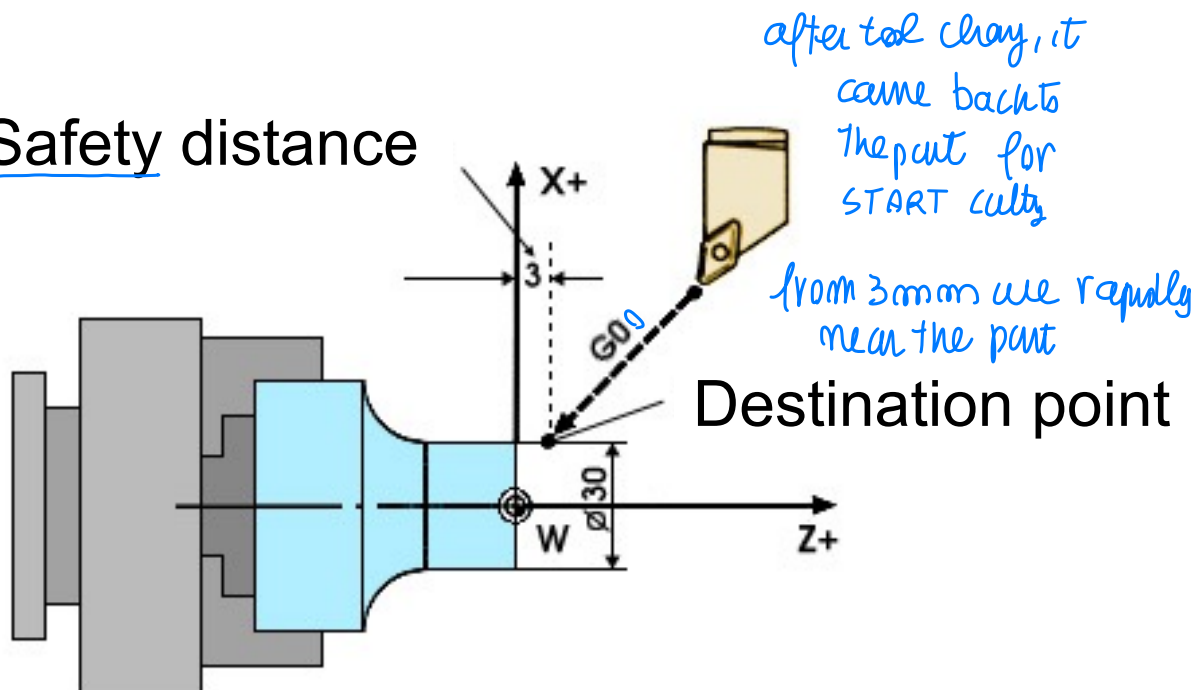
Different rapid rate for each axis



EXAMPLE OF POSITIONING

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Safety distance



RAPID MOTION at Z=3

N100 (G00) X30 Z3

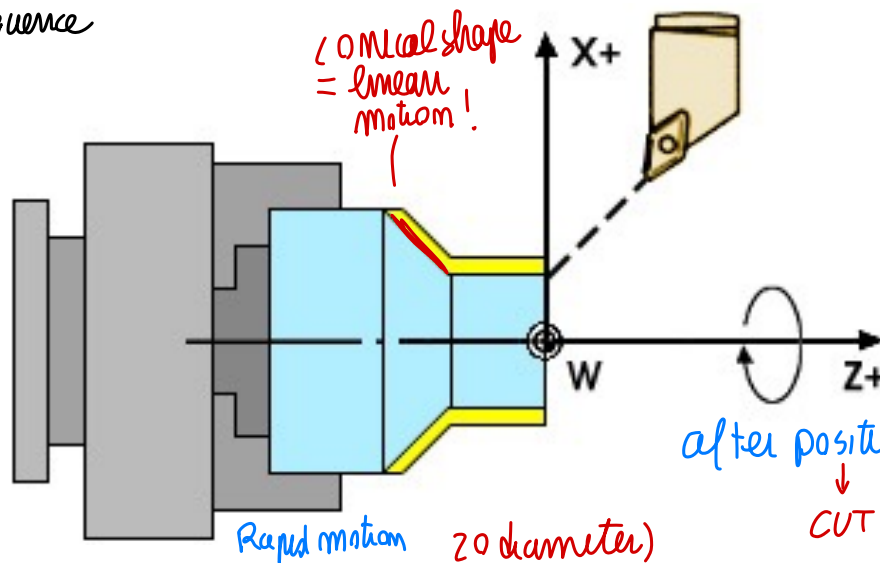
position @ Diameter 30



LINEAR INTERPOLATION (G01)

NOT DISCONTINUITY, all
executed well in sequence

When moving
along conical shape
→ cutting speed changes!



during conical shape machine
we move two axis
simultaneously to guarantee
the linear path, we need
to specify how to keep axis
velocity (*)

after positioning --

CUT the material

profile as set of
line (linear interpolation)

0.2 mm/REV = f

move the tool and
cut material during
motion

N180 G00 X20 Z2

N190 G01 Z-15 (F0.2)

N200 G01 X40 Z-25 (*)

through
absolute
coordinates!

continuous smooth
movement, operations
executed in sequence

move the
tool with defined given motion

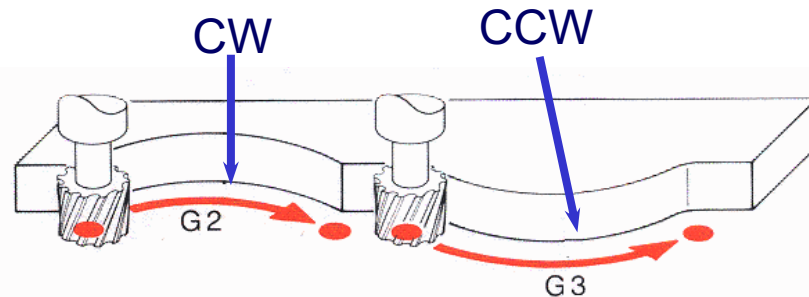
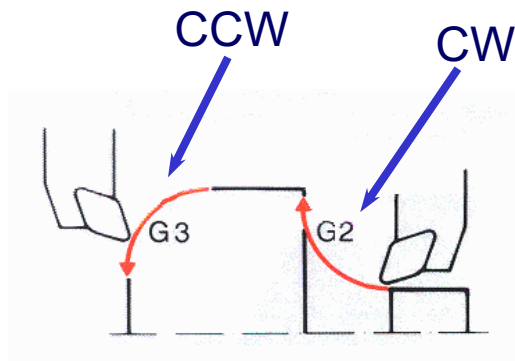
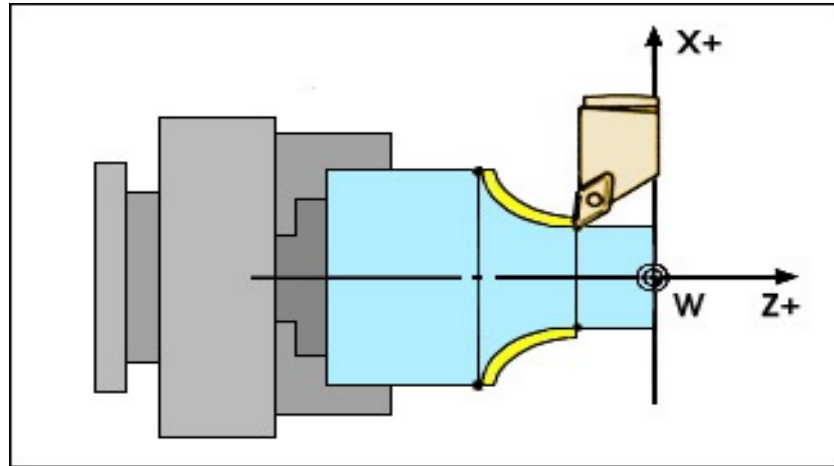


CIRCULAR INTERPOLATION

{ control system has to manage the motion -- }

to create arcs

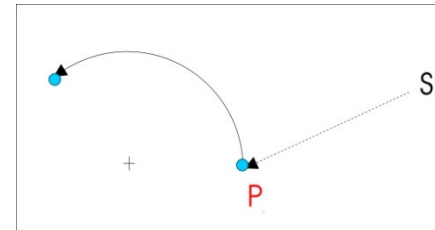
- G02 CW
 - G03 CCW
- give the final point
↳



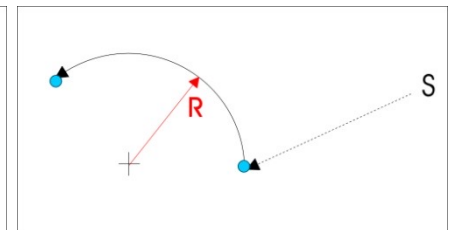
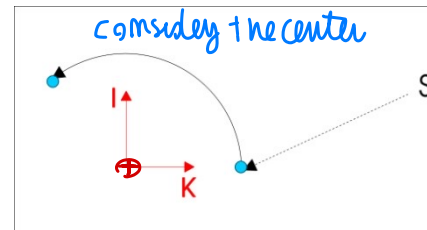


1. Initial point coordinates

2. Choice between: *specification of that to define the circular motion*

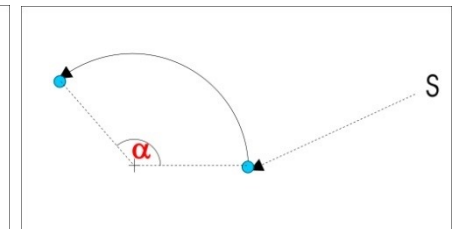
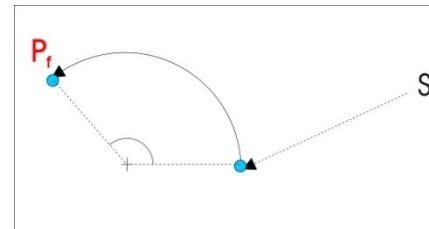


- Center coordinates
plane x,y (G17): I,J
plane x,z (G18): I,K $\Rightarrow$
plane y,z (G19): J,K
- Radius R



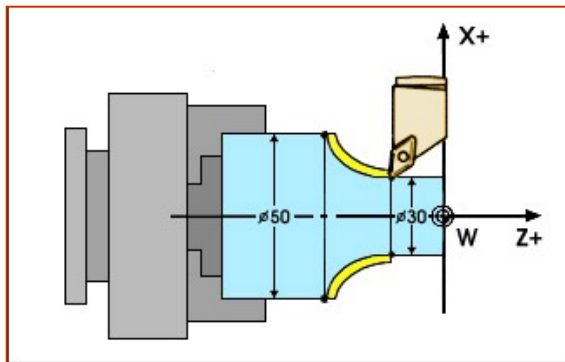
3. Choice between:

- Final point
- Angle range

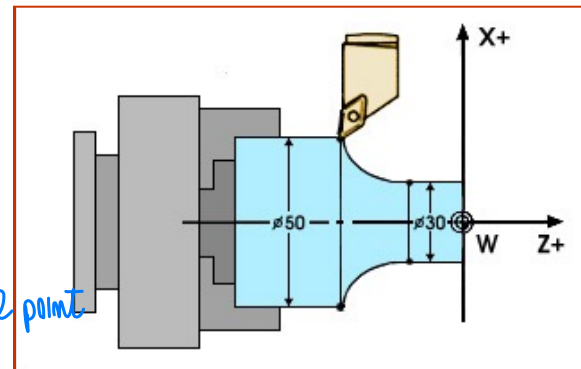




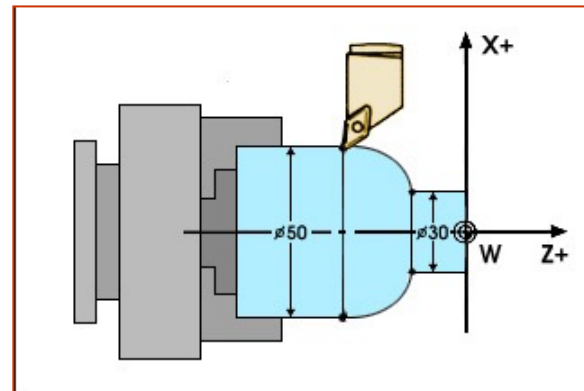
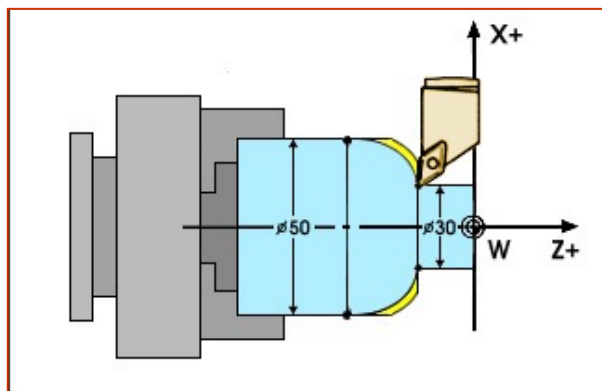
CIRCULAR INTERPOLATION: EXAMPLES



• COUNTER
CLOCKWISE
MOTION
• giving final point



N100 **G02** X50 Z-35 **I150 K-25**
center & z position



90° arc angle!

N150 **G03** X50 Z-35 **I30 K-35**
counter
clockwise
same
end point

arc center position

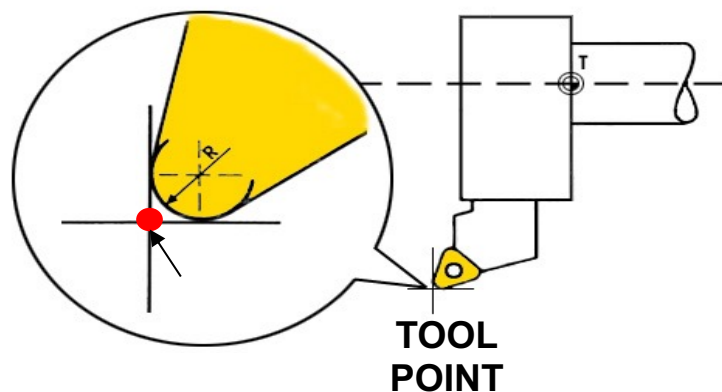


TOOL PATH

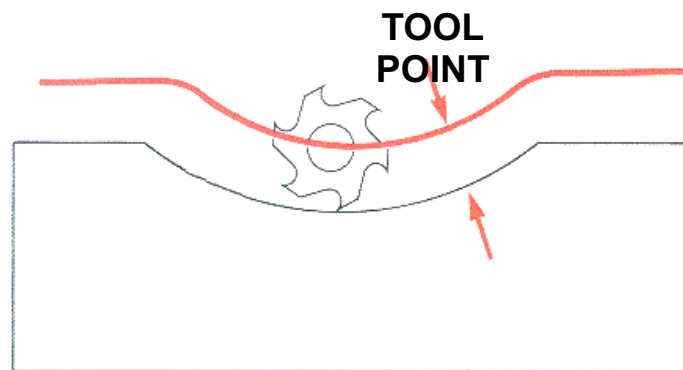
we move Ref. point
↓ referring to a **TIP** well defined point

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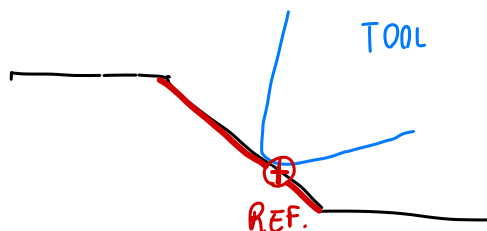
The tool path is programmed referring to a specific tool point.



move point along trajectory well defined --



to have tool cutting properly



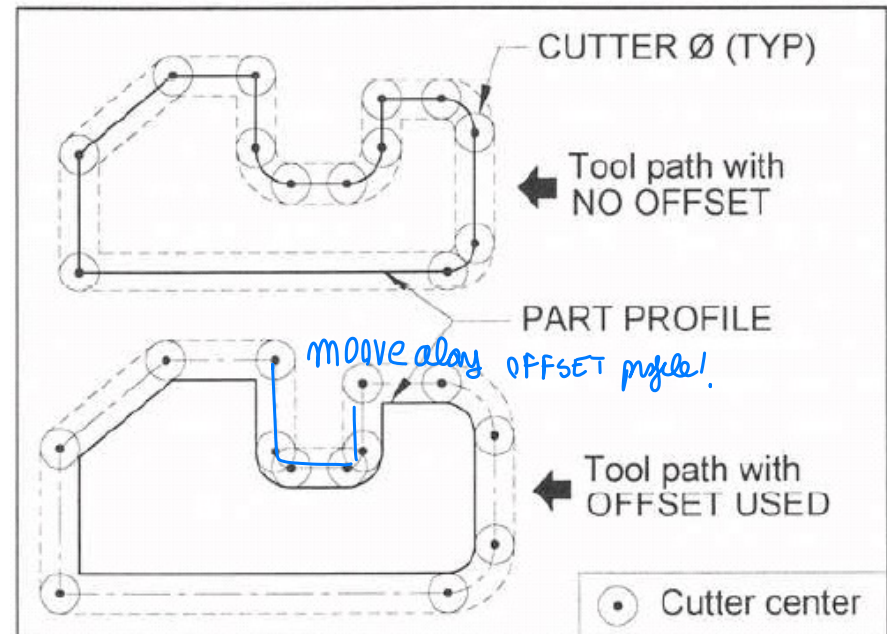
respect trajectory → you need to compensate the position
(easy to understand in a milling operation) →



In several machining operations we have to make distinction between:

- tool path
- distances in the technical drawing

The tool compensation (or cutter radius offset) helps to program using the information in the technical drawing.



(IF NOT COMPENSATE → we obtain a non correct final profile → compensate with a radius offset);
↓
We have to compute the OFFSET



TOOL COMPENSATION

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The NC calculates the trajectory of the tool point on the basis of the milling tool radius R .

OFFSET
can exist or NOT!
↓
sometimes in MILLING
we don't care about
OFFSET →

PATH define how to move!

Stay on the right

	G41 LEFT	
	G42 RIGHT	
	G40 - NONE -	

OFFSET related
to way of motion

● basic
compensation
INFO

IF plane to cut,
we don't care
about offset, just
move on surface
(phase mully of plan on surface) → NOT needed
compensation!

instead for profile you
need offset according
to motion

That kind of programming is TIME CONSUMING also for simple situations!

a lot of INFORMATION to give!

↳ TO IMPROVE PROGRAMMING we can
use CAD Model

↳

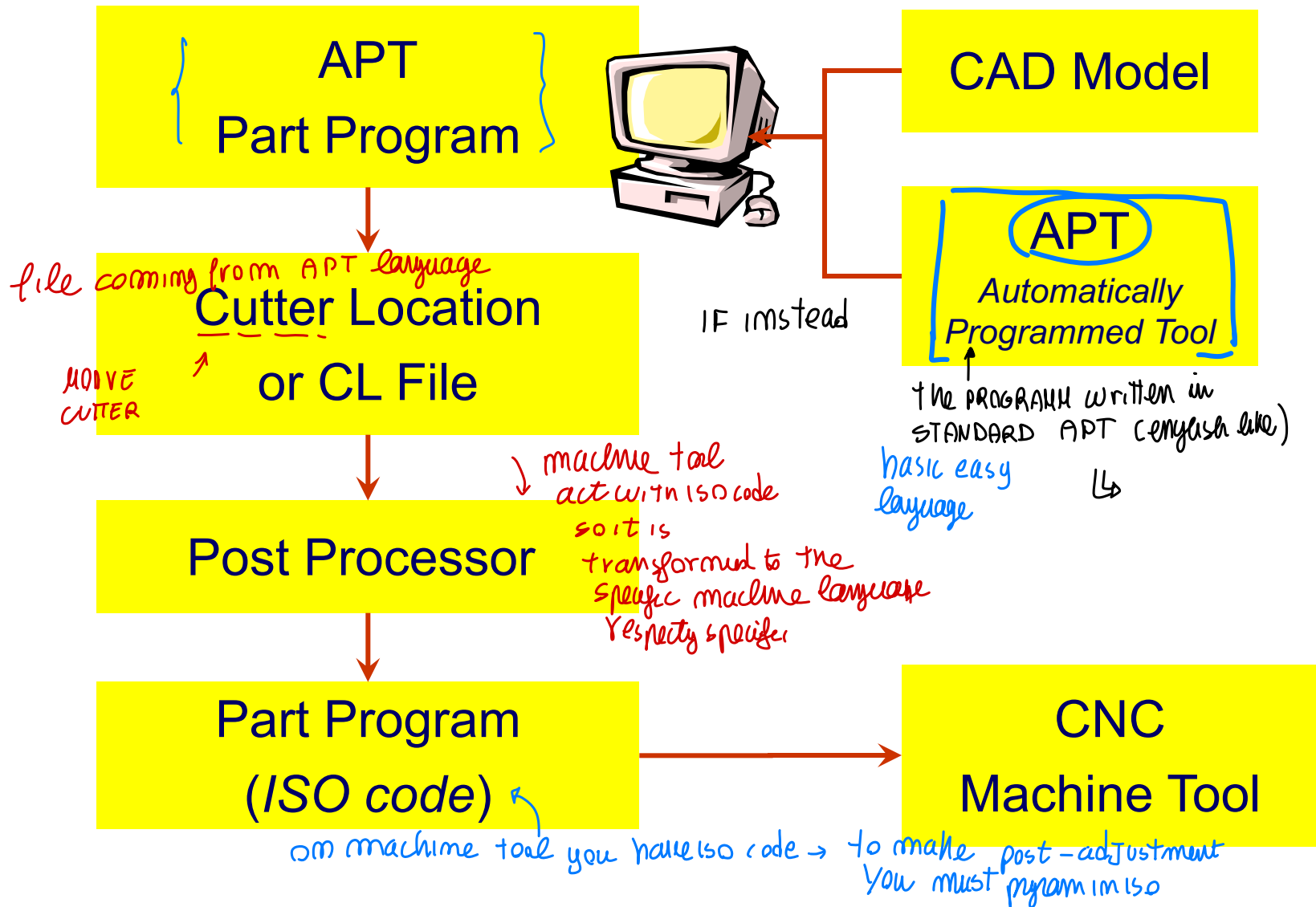


CNC PROGRAMMING

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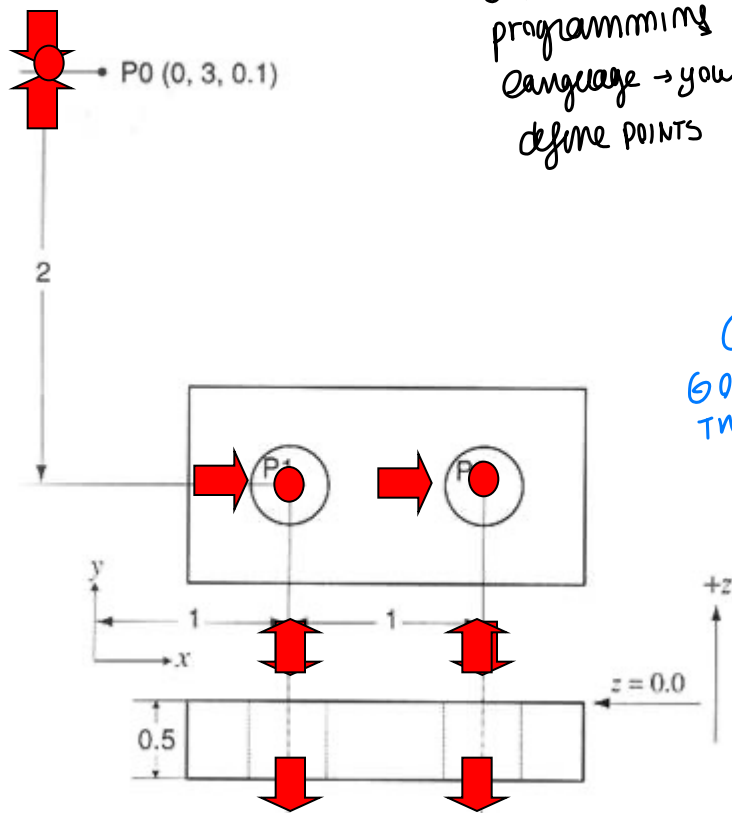
manually programming is complicated!
(You need to know instructions)

FIRST EVOLUTION *easy the PROGRAMMING!*





DRILLING



you START define the geometry

like normal programming language → you define POINTS

Go delta, Go through the hole

minimal geometry

```
P0=POINT/0.0,3.0,0.1
P1=POINT/1.0,1.0,0.1
P2=POINT/2.0,1.0,0.1
```

FROM/P0 start point

GOTO/P1 where to go, without specific INTERP.

```
GODLTA/0,0,-0.7
```

```
GODLTA/0,0,0.7
```

GO BACK to come back to P1

```
GOTO/P2
```

```
GODLTA/0,0,-0.7
```

```
GODLTA/0,0,0.7
```

same

```
GOTO/P0
```

Return to start position

kind of language more understandable (evolution of programming)



Other specs:

- MACHIN/ name of the machine tool
- COOLNT/ coolant on/off
- FEDRAT/ feed rate
- SPINDL/ rpm
- TOOLNO/ tool number
- CUTTER/ tool compensation
- INTOL/ o OUTTOL/ tolerance
- END (program end)

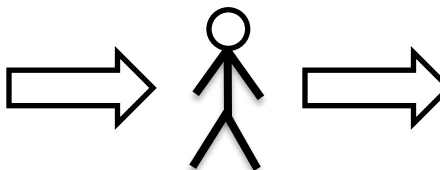
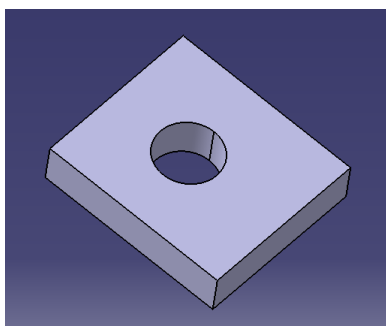
*simple programming
language!*



Usually, the **creation/modification of a part program** can be done in two ways:

- manually on the machine;
- using a Computer Aided Manufacturing software.

The programming on the **machine** is used when the part to be produced is simple.



ISO Code

```
...  
G01 X0.0 Y0.0;  
G01 X2.0 Y0.0;  
G01 X2.0 Y2.0;  
G01 X0.0 Y2.0;  
G01 X0.0 Y0.0;  
...
```



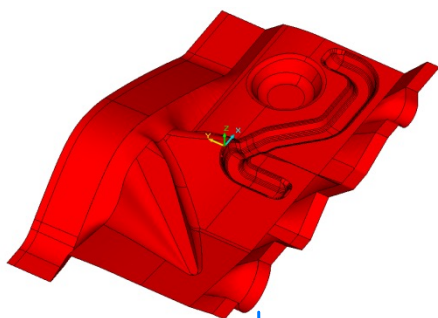
CNC PROGRAMMING: CAD/CAM

The usage of a **CAM software** happens when the part is hard to manufacture, it has complicate features or requires the interaction of multiple programmers.

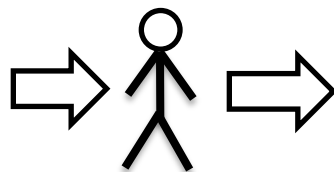
(simulate any issue about collision and so on...)

simulate TOOL motion, path ON WORKING ENVELOP

use the TOOL to plan machining solve the coding issue -- define traj through TOOL

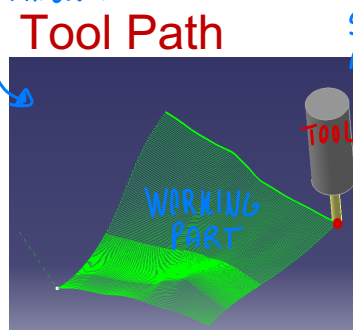


solid model starting point



It can be easy to have a sw that enable you to chose the path, position the TOOL and chose it (give final geom. & tool) define WORK. parameter

Carefull to collision ISSUE



simulate machining

enables to OUTPUT CUTTER LOCATION FILE

APT Code

OUTPUT

```
PARTNO / APT-1
CLPRNT
UNITS / MM
NOPOST
CUTTER / 10.0
$$$GEOMETRY DEFINITION
SETPT = POINT / 0.0, 0.0, 0.0
STRPT = POINT / 70,70,0
P1 = POINT / 50, 50,
```

ISO Code

...		
G01	X0.0	Y0.0;
G01	X2.0	Y0.0;
G01	X2.0	Y2.0;
G01	X0.0	Y2.0;
G01	X0.0	Y0.0;
...		

at the end, different post-processor that create iso code from by input mach

Post processor



3D machining from a block PROGRAMMING that in ISO code is impossible!

Roughing (finish) we have a lot of different motion / tool changing / rotation of pat --
HAND PROGRAMMING is almost impossible

(LOT of behind computation)

<https://youtu.be/8FGpvWqLK-s>



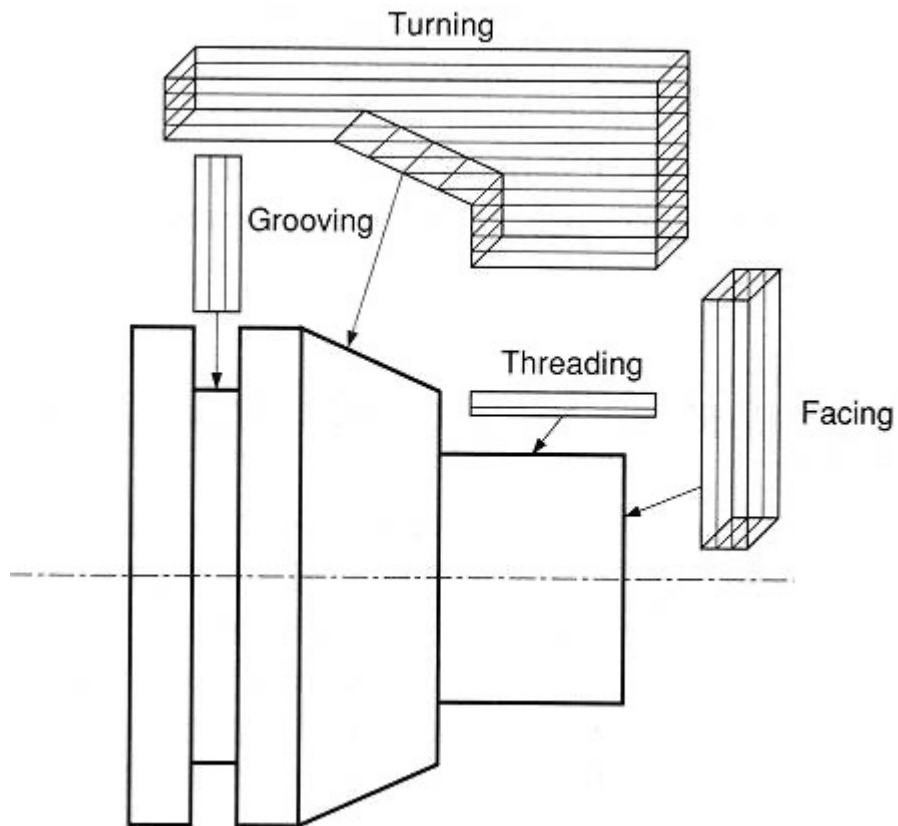
CAM: TOOL PATH

in **PATH PROGRAMMING**

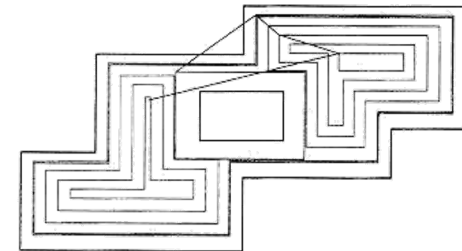
in simple situation, sw that enable to specify easy the profile by the geometrical -

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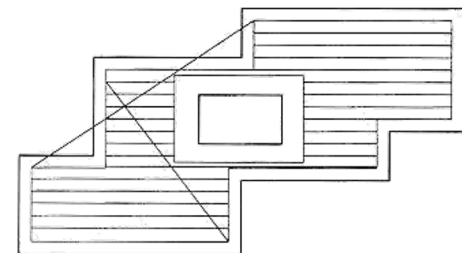
Turning



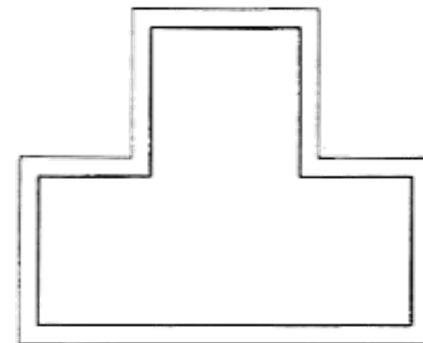
Milling



(a)



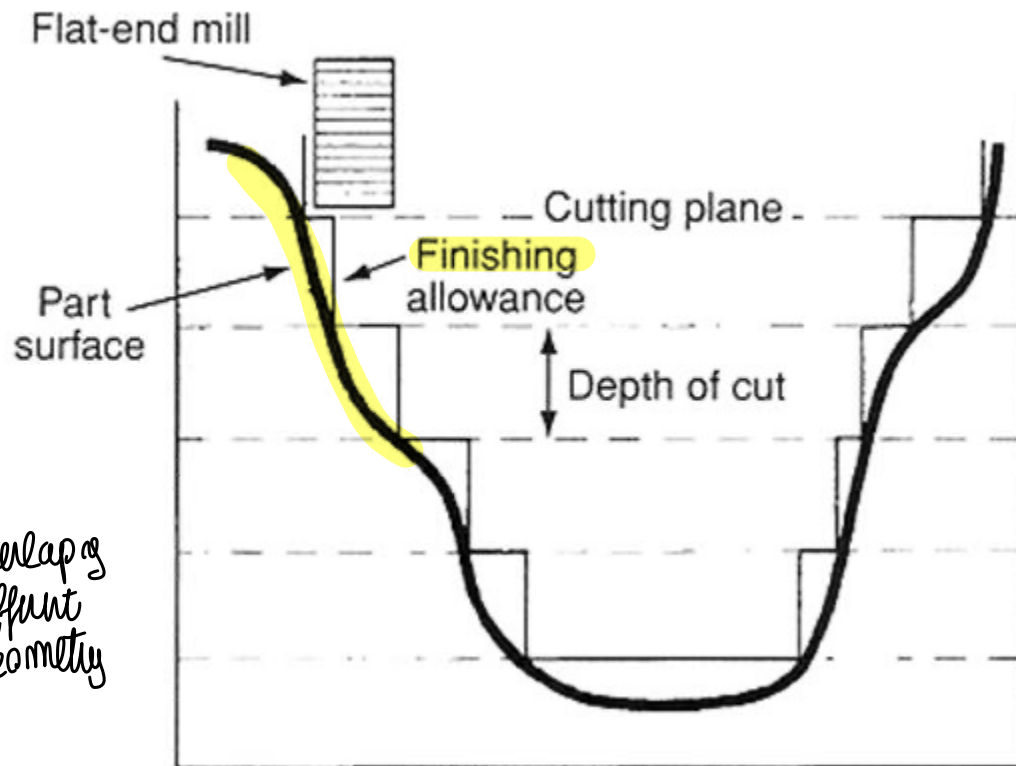
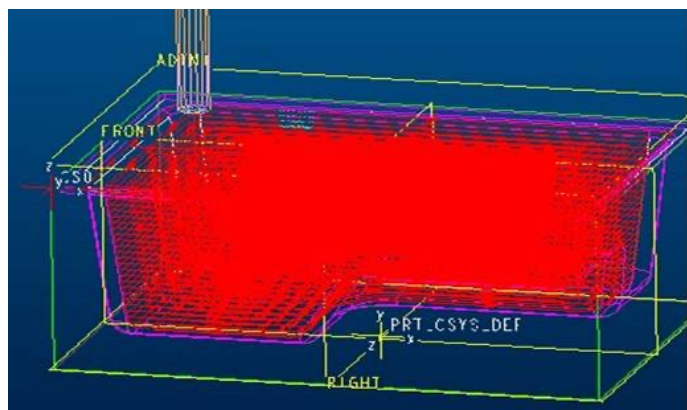
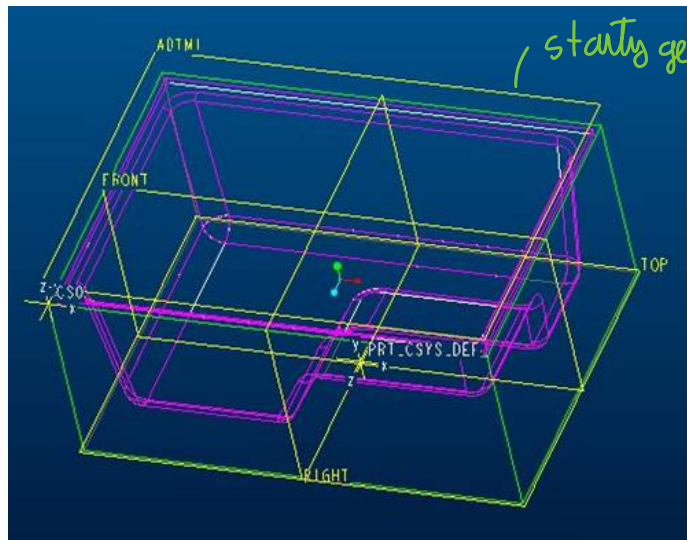
(b)



(c)



POCKET definition through CAM SOFTWARE

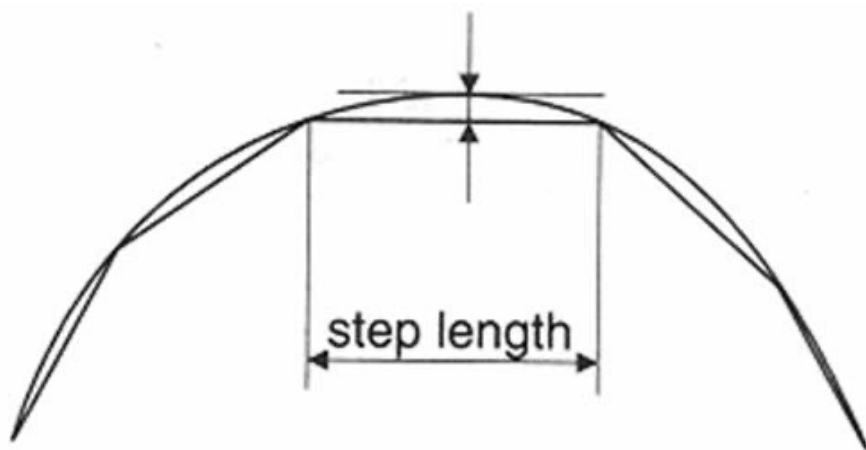


+ Roughing operations (3AXIS)

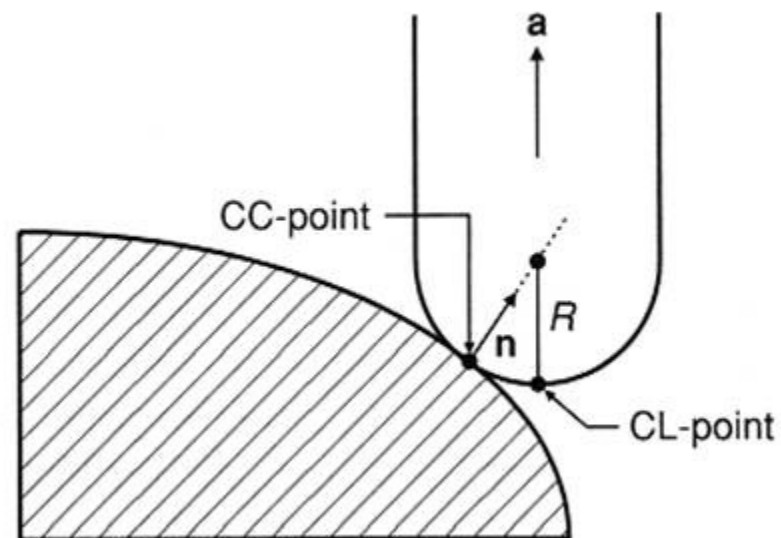


using interpolation, you have approx
errors on traj!

Tool path approximation



Tool compensation

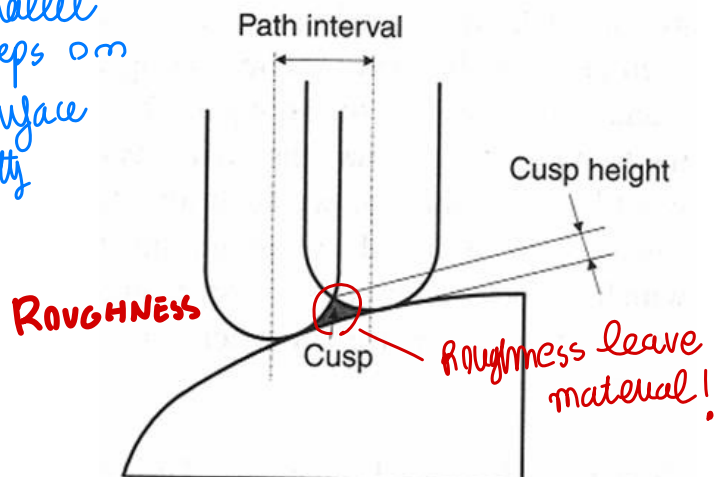


we need to compensate
properly respect TOOL shape, to
have right CONTACT



Problems in tool path generation

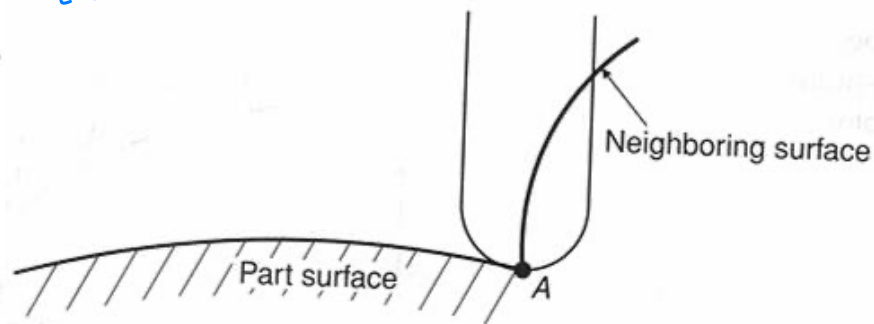
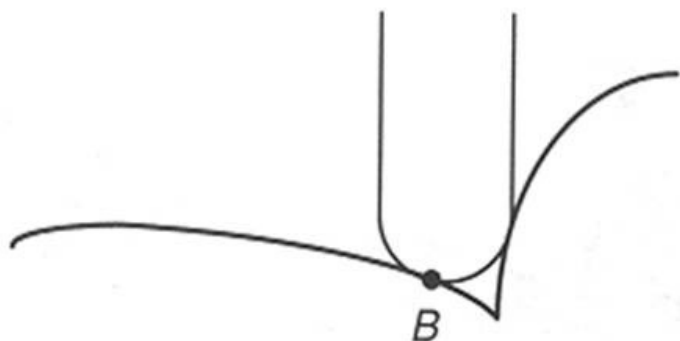
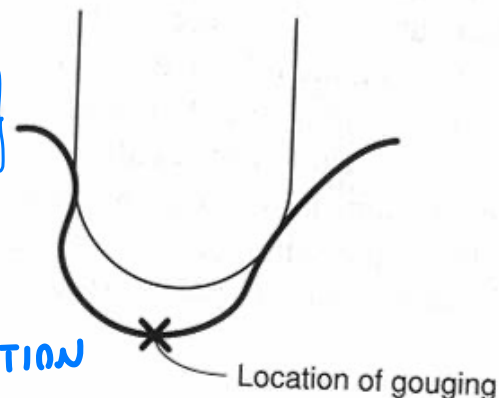
Parallel
steps on
surface
cutting



{ to create some profile that
tool is NOT correct! }

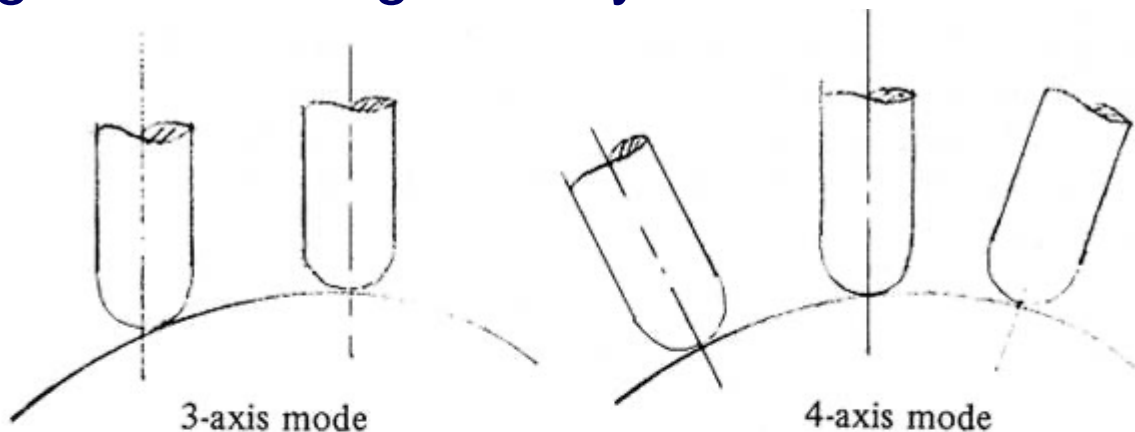
{ shape profile
problems! }

VERIFICATION
to do in
CAM SW



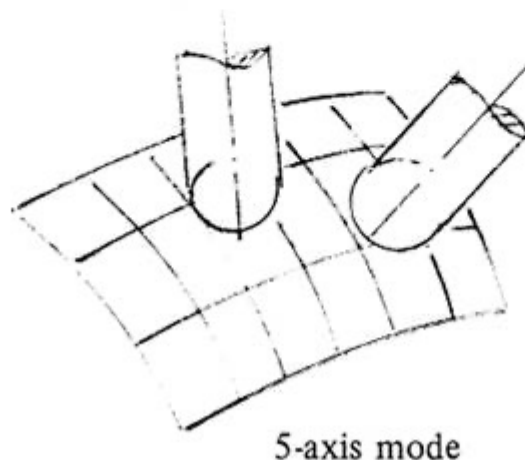


Milling of freeform geometry



While moving the tool ...
axis parallel to ITSELF

We can use more axis
motion



to correctly cut a surface we
need to move multiple axis of
tool (also ROTATION)

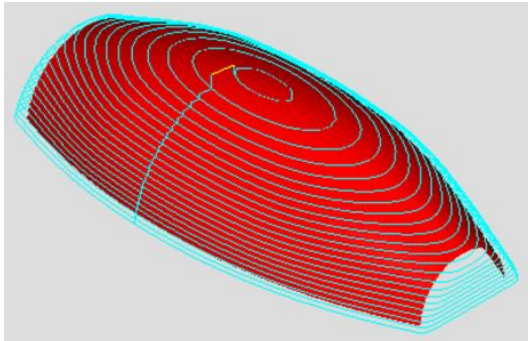
X Y Z A B → so many
motion comment



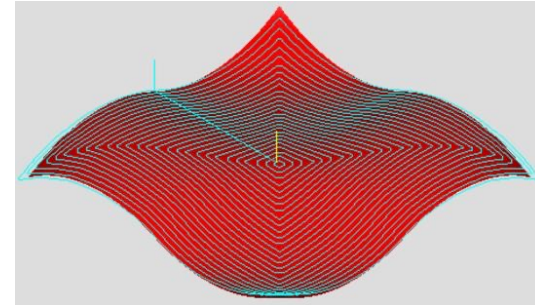
CAM: TOOL PATH

different path alternatives

42

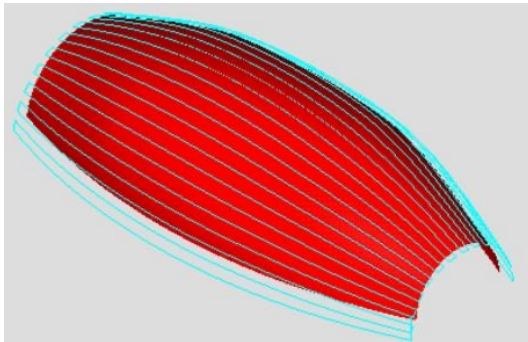


Z level
curves
*cutting along
parallel plane*

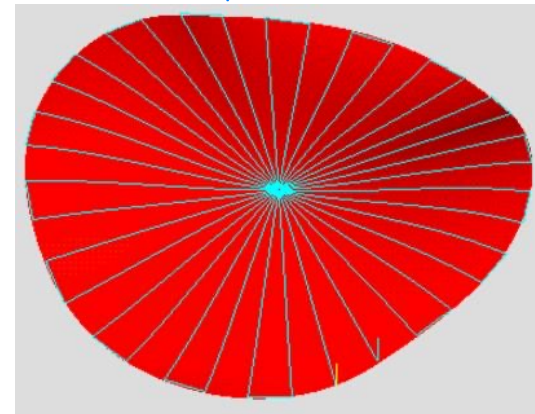


spiral path

Other complex path

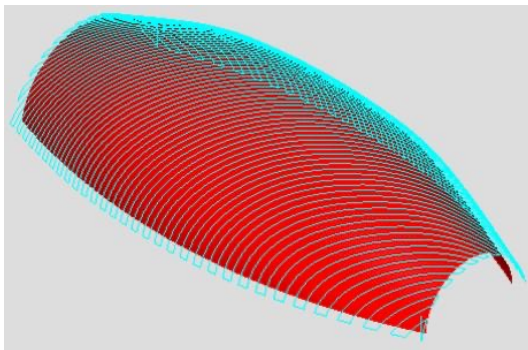


constant
parameter
curves *function
of 2 parameters --
better cutting! generated
changing one
parameter a time*



radial path

*FIND the best one for our purpose
→ best surface*



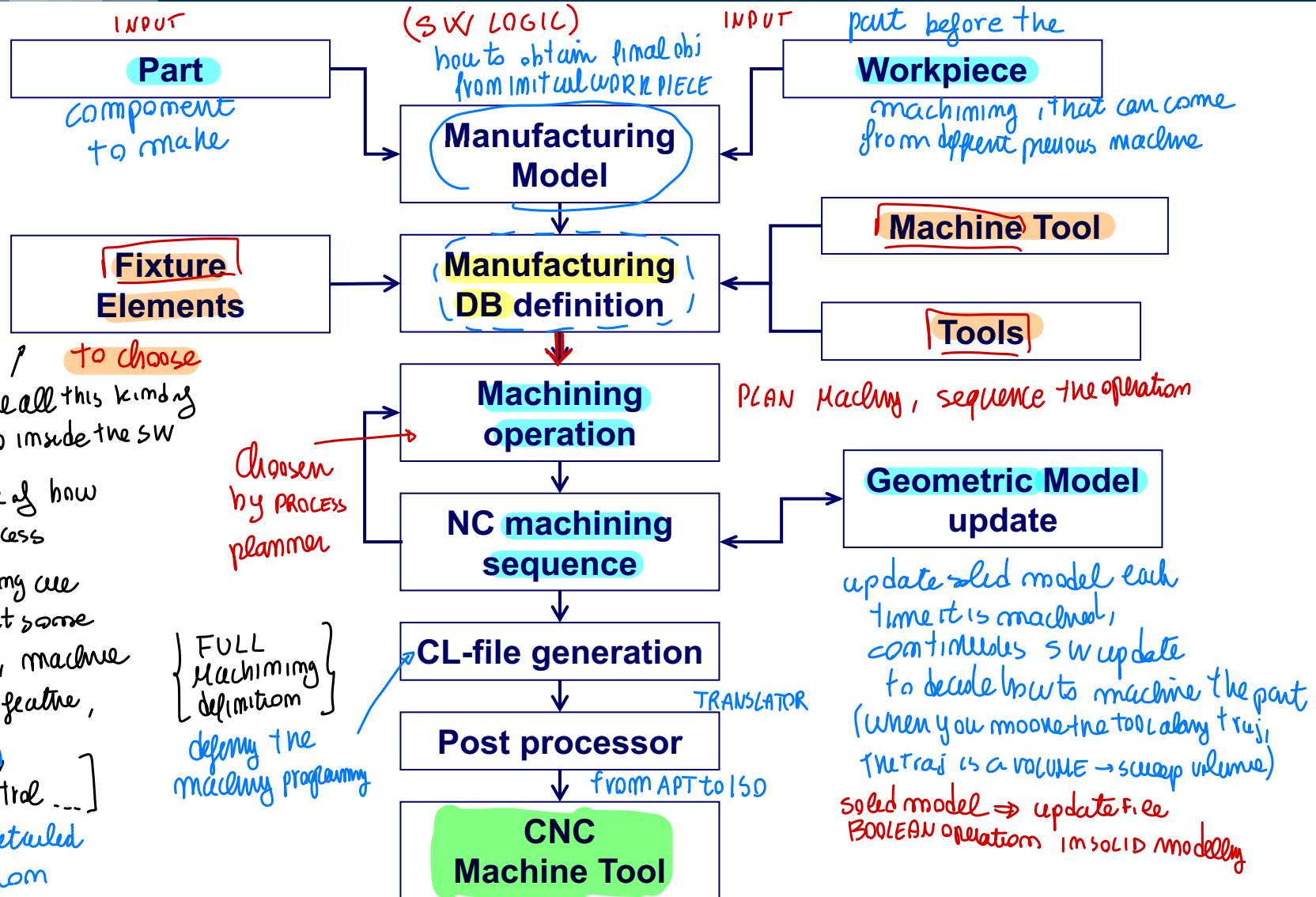
parallel
plane
curves
*↓
different direction
parallel planeset*



CAD/CAM

that SW has a LOGIC schema of what we can do...
applied for any SW type

43





STEP-NC (ISO 14649)

That slide 43 situation is the CURRENT usage!

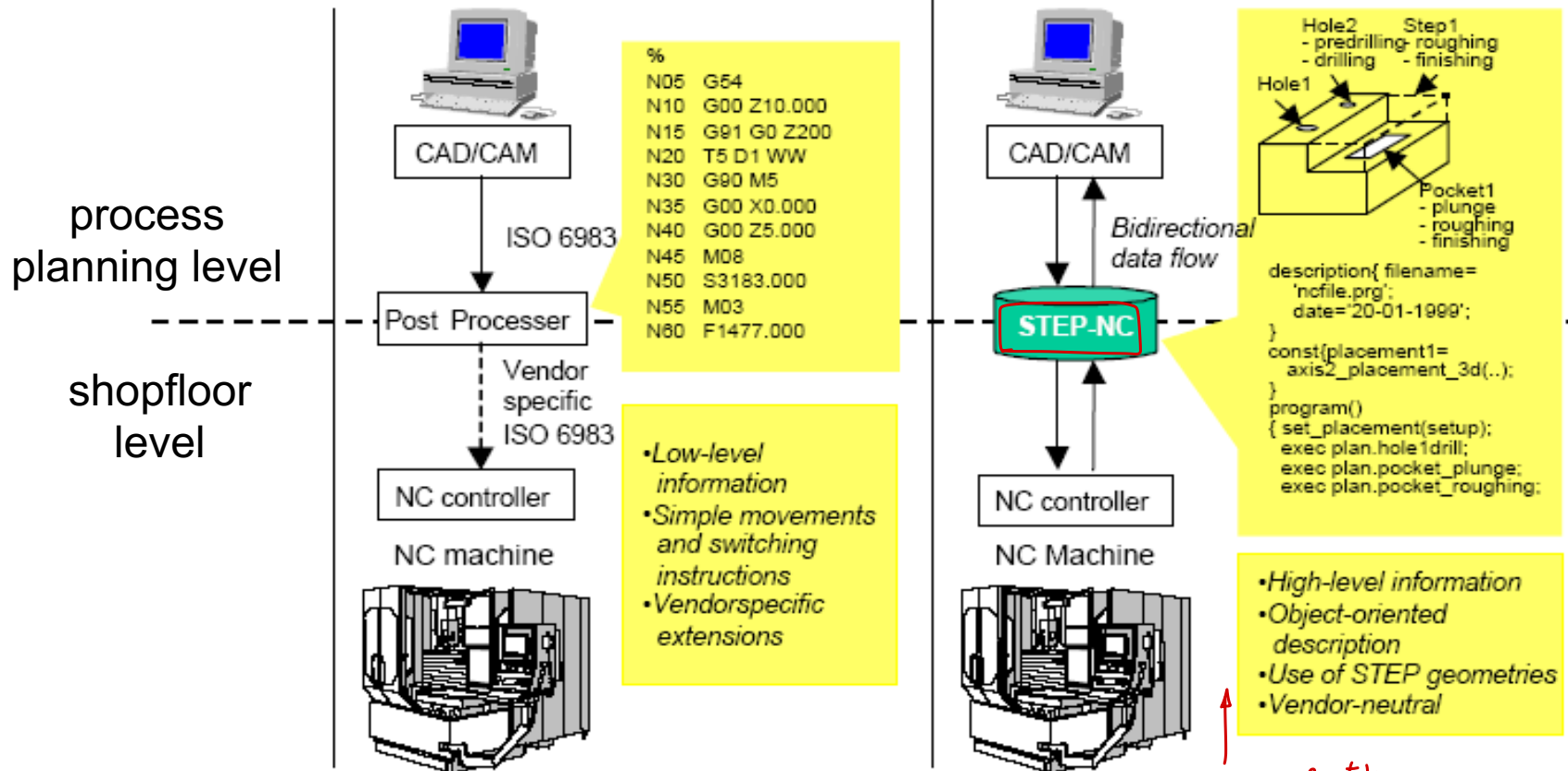
44

We have another possible program definition

current situation

UNT50 spread

new interface



ISO step-NC standard
high level WORKING without
going low level definition!

machining through geom
features definition
(feature level PROGRAM SW) → geom feature related to machine

new way -- a SW help the user in
PROCESS PLANNING (how to machine the part)
TEST different possibility



STEP-NC (ISO 14649)

→ describe operation and geom features of the part → HIGH LEVEL model

45

↓
LOW level managed by CNC machine
path programming managed by machine

STEP-NC changes the way that manufacturing is done by defining data as "**working steps**": **a library of specific operations that might be performed on a CNC machine tool.**

In other words, it breaks down every machining operation into the steps required to perform the operation.

In the past, CNC machine tools had to be programmed using G and M codes (from ISO 6983), which were instructions that only told the machine what moves to make, without any semantic content that referred to the part being processed.

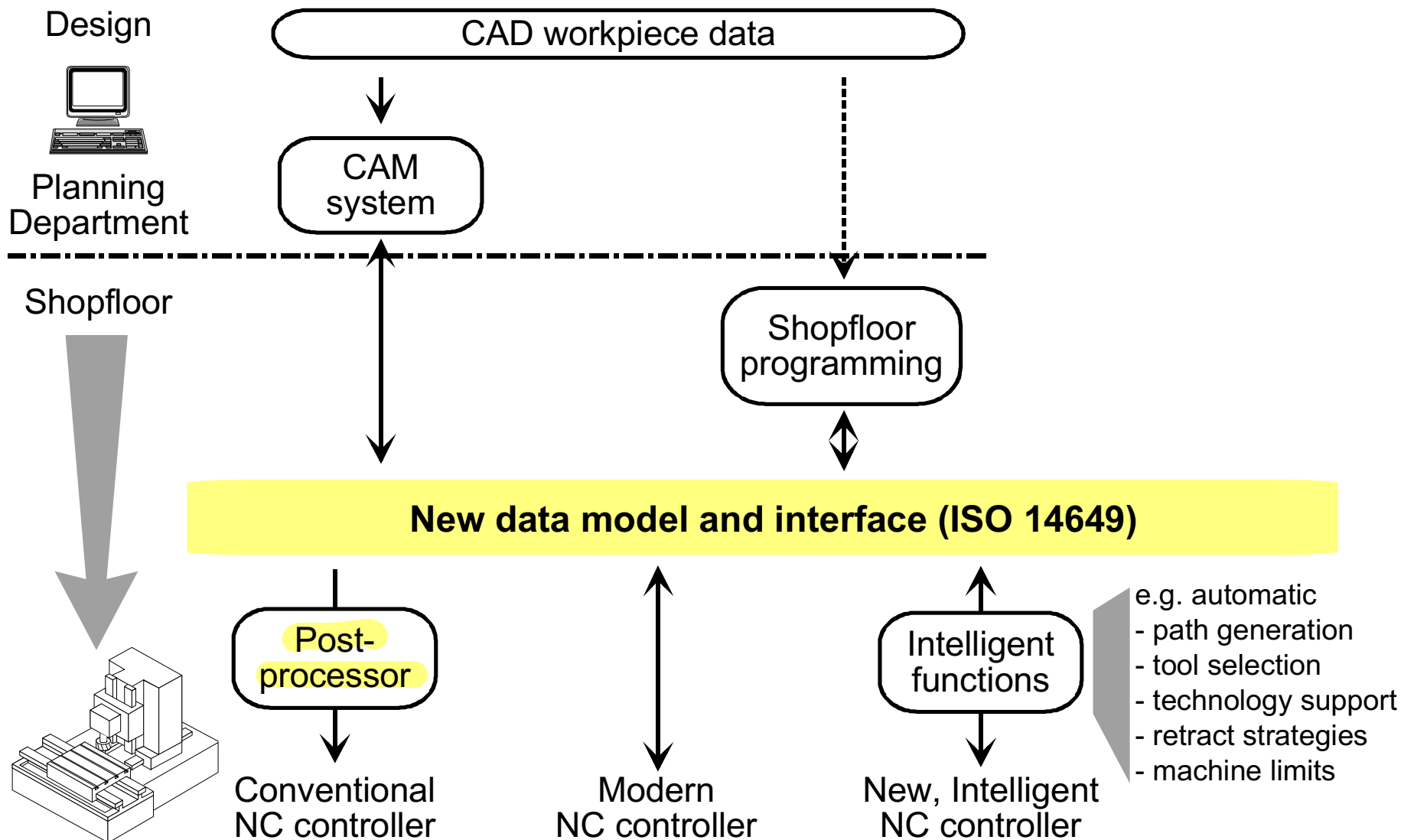


STEP-NC (ISO 14649)

You may have POST-PROCESSING..

the ISSUE is to define intelligent control syst that choice for you!

46

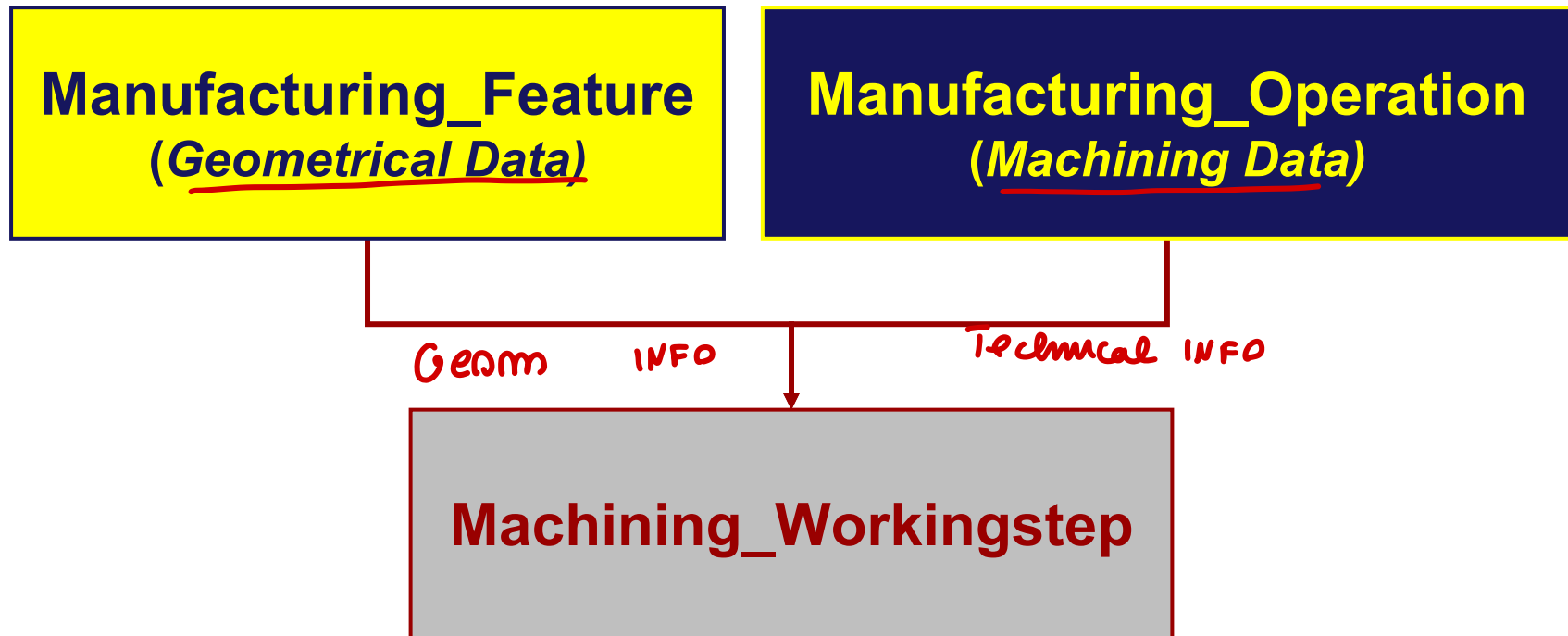




STEP-NC (ISO 14649)

how + hat high level language works?

47



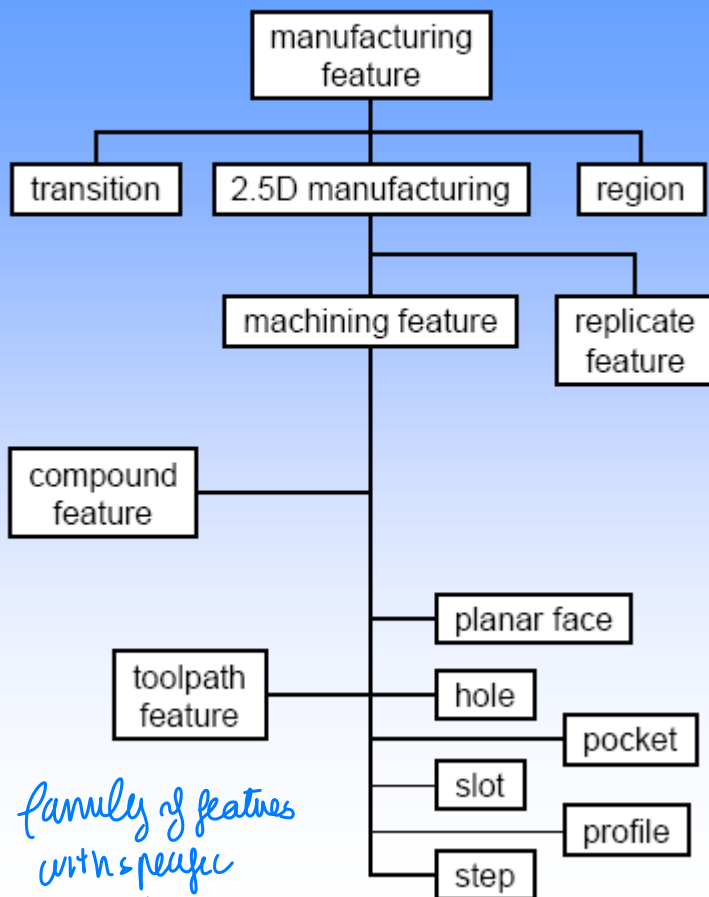


STEP-NC (ISO 14649)

48

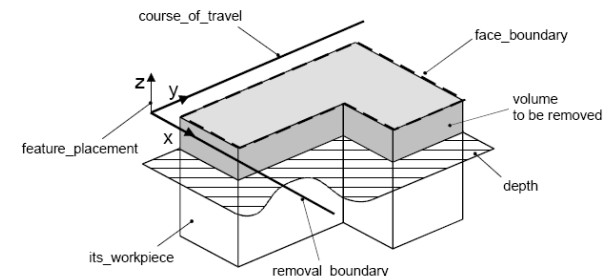
[EXAMPLE]

STEP-NC Features

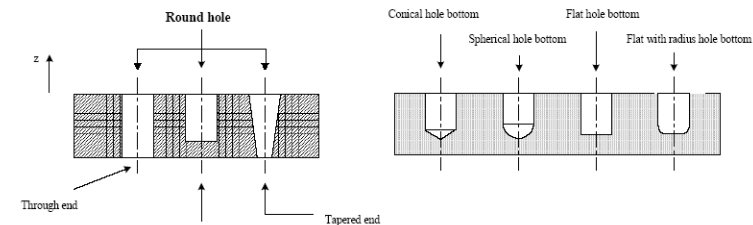


family of features
with specific
geometry

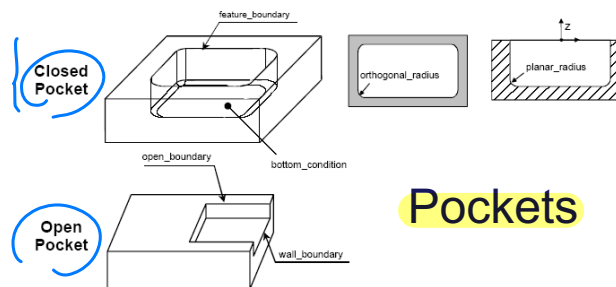
Planar Face



Round Holes



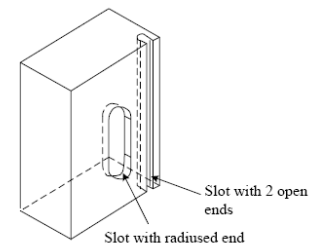
Hole Taper and Bottom Conditions



Pockets

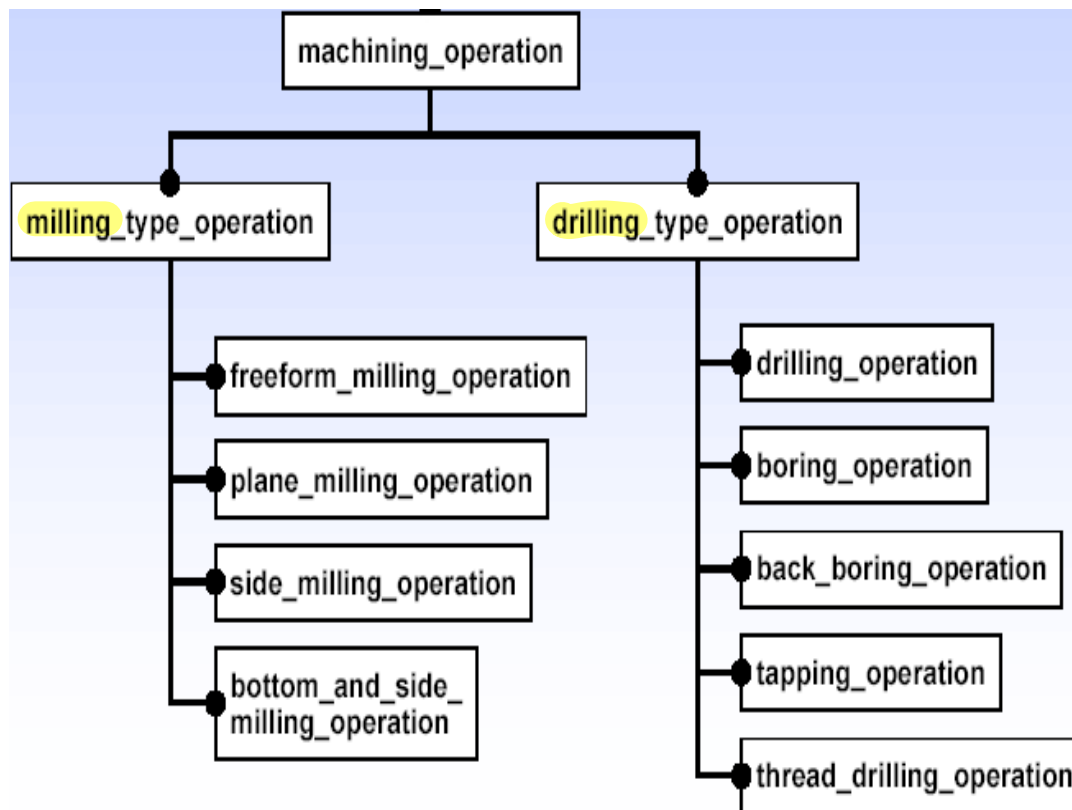
many different features
definition!

Slots





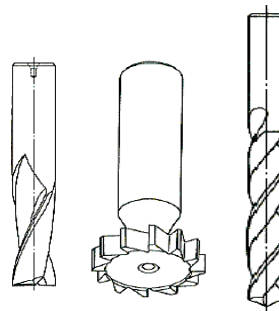
WORKING-STEP definition → kind of operations



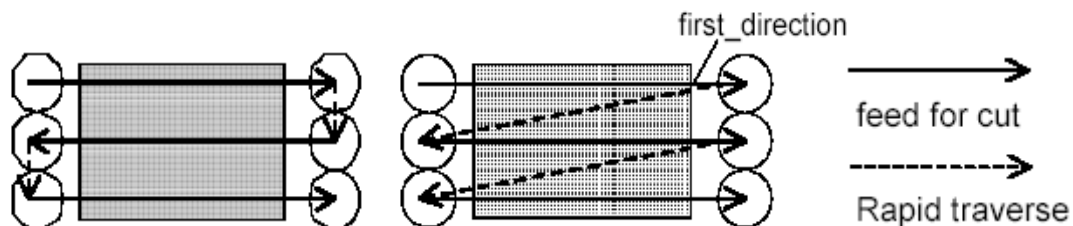


Machining Operation

Tool Requirement



Machining Strategy



Technology Parameters

Spindle Speed [RPM], Feed [mm/min]

Machine Functions

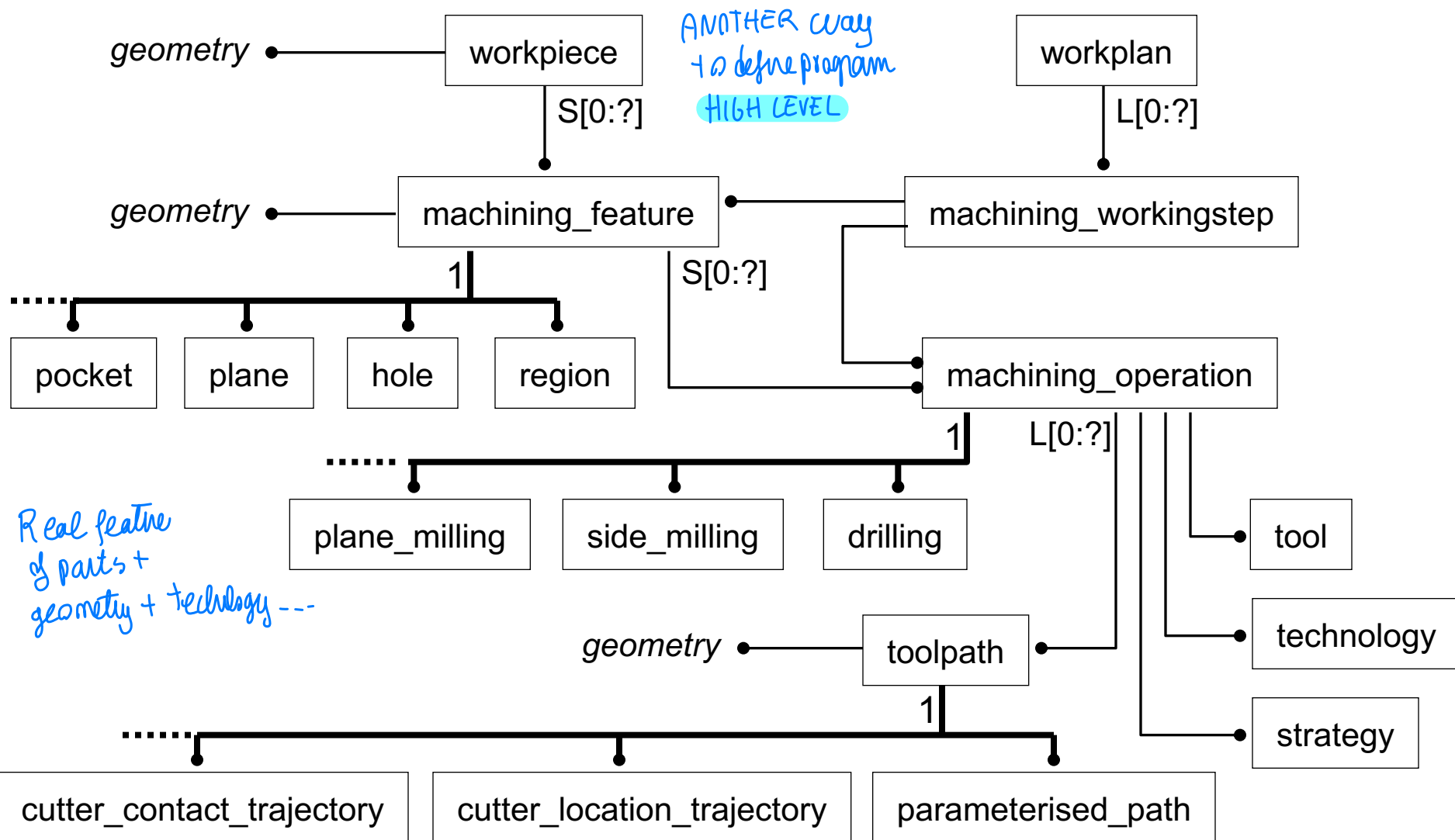
M Functions: Coolant ON/OFF.....



STEP-NC (ISO 14649)

FINAL WORKPLAN Schema
set of WORKING STEP machining

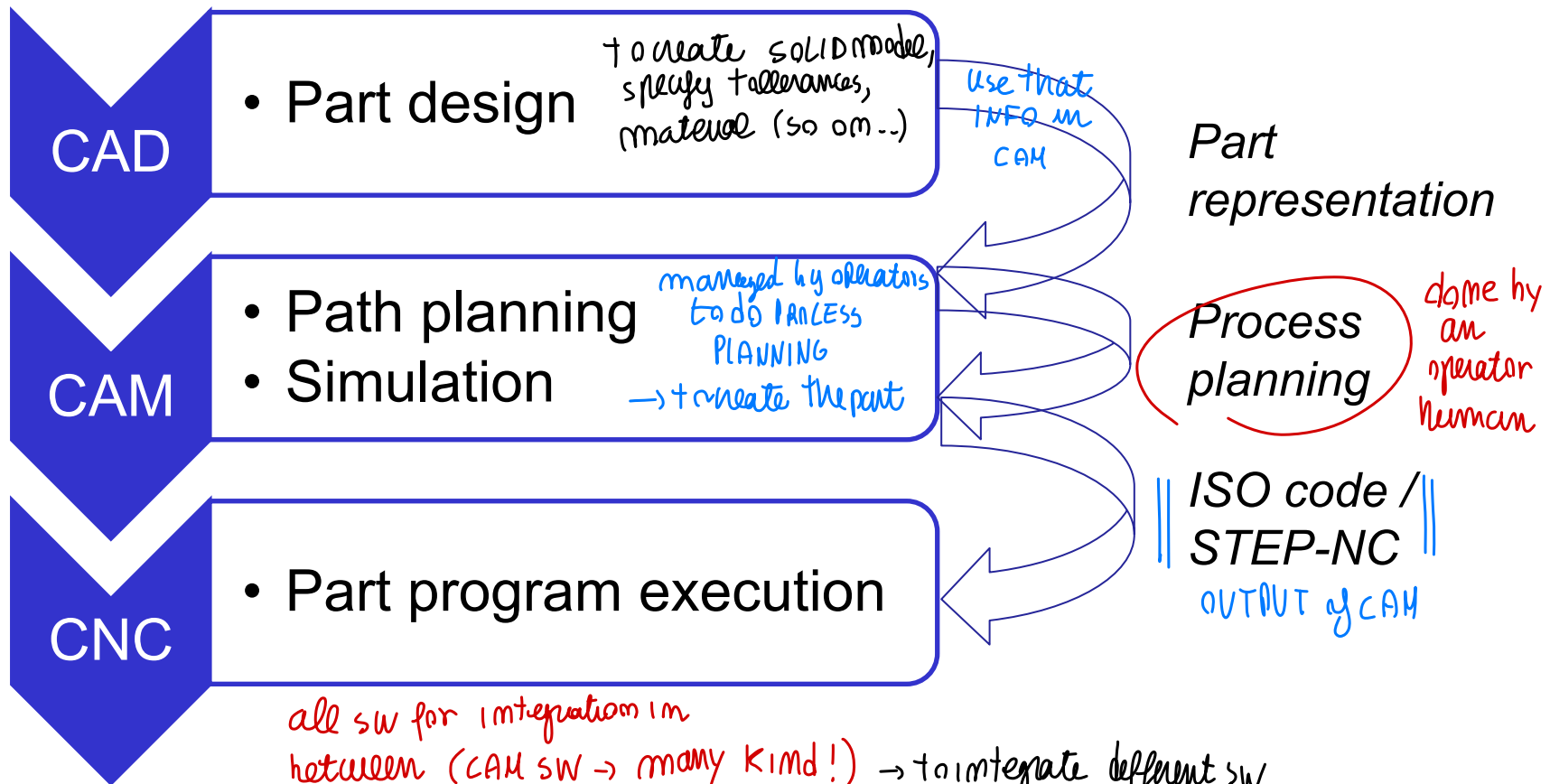
51



Advantage of this only from compatible CNC → that can directly obtain STEP-NC file



↑
small portion of the integration



all sw for integration in

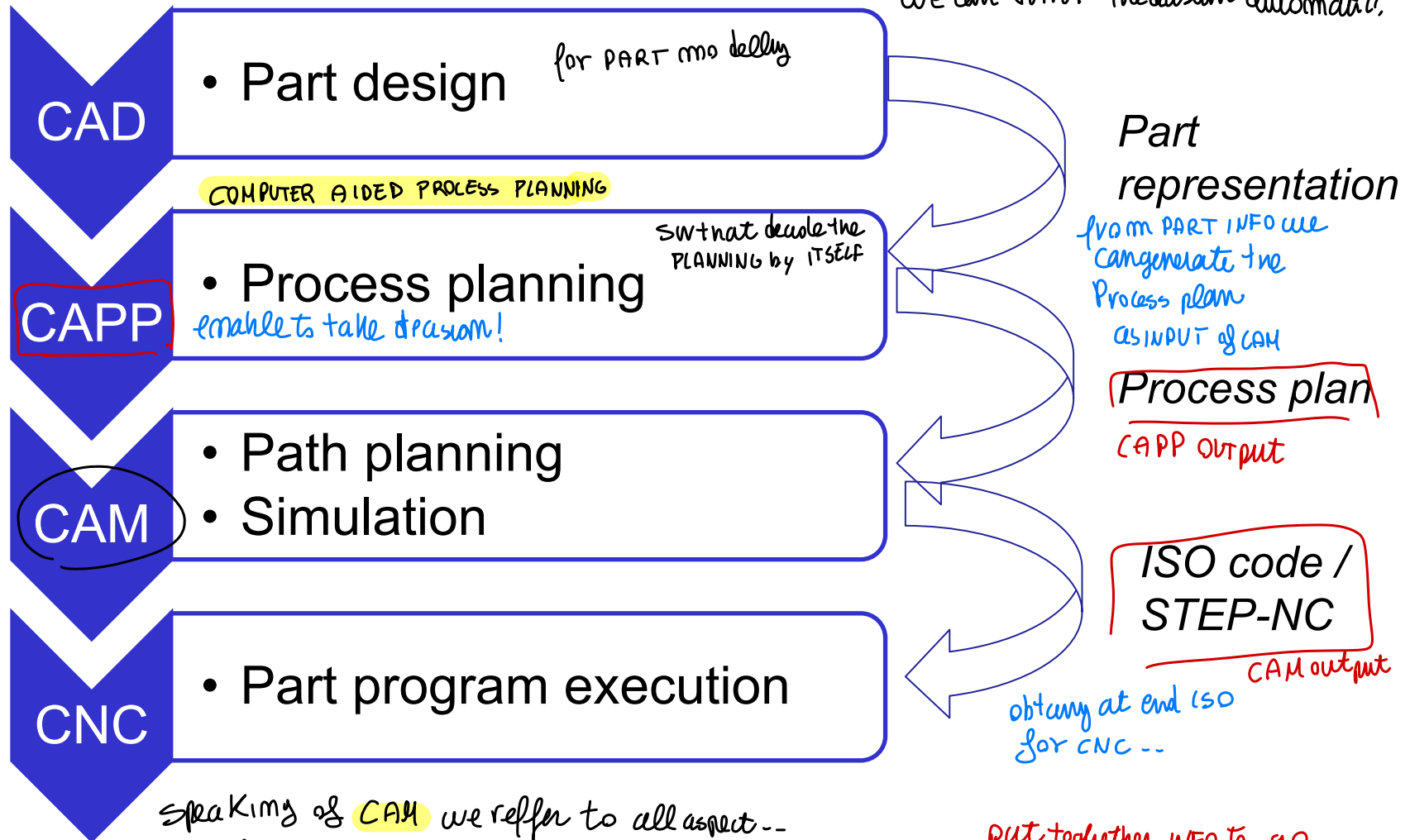
between (CAM sw → many kind!)

*→ to integrate different sw
You need compatibility issue*

specific integration

needed! → iso code after CAM

*{ usually we need an operator to }
do PROCESS PLANNING!*



Why to have an operator for process plan?
We can output the decision automatically,

Speaking of **CAM** we refer to all aspect --
while CAM → related to use computer to take decision →

Optimize Process Plan by SW help

put together INFO to go
from design to machine without
human work